Measurement of Occupancy and Crowding in Embedding Spaces

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Abstract

Embedding-based systems — retrieval, retrieval-augmented generation, deduplication, clustering — depend on a single geometric property: items that must be distinguished must remain distinguishable under the similarity metric. We call the failure of this property *crowding*, and we argue it is the correct diagnostic target for embedded corpora: crowding is a loss of *resolution*, it is predominantly *local*, and it hides from the global averages (mean pairwise cosine, isotropy scores, effective rank) that practitioners routinely consult. This report describes `ambit`, an open-source instrument that measures how a corpus occupies its embedding space. The design is organized around three commitments. First, occupancy is measured *continuously over all scales*: the primary object is the empirical pair-distance distribution, read exactly (no histogram bins, grid cells, or fixed neighbor counts, whose size and placement provably change the answer), against analytic or cheaply simulated null models. Second, every statistic is *null-calibrated*: values are reported as deviations from a matched well-spread reference, including an anisotropy-conditioned reference that separates the ubiquitous "cone effect" from clustering the cone cannot explain. Third, diagnosis descends to *named entities*: per-item scores (a distance-to-measure field with Wasserstein-stability guarantees; expected retrieval collisions under an exact Gaussian query-noise law) replace per-cell counts, which cannot identify the items at risk. We derive an operational scalar, the *resolution bandwidth* σ^* — the largest query-noise scale at which the expected number of retrieval collisions per item stays below one — prove its conservative (union-bound) semantics, and show that the widely used uniformity loss of Wang and Isola is the logarithm of a Chernoff bound on the same quantity, endowing its temperature parameter with units. Controlled experiments quantify the fragility of binned occupancy (a half-cell lattice shift moves a hexbin peak count by 30% and relocates the peak on fixed data), verify the confusion law to Monte-Carlo precision (maximum deviation 9×10^{-4} over a 6×4 grid of angles and noise scales), and trace a planted 5% near-duplicate pocket through every layer of the instrument: whole-curve rejection of uniformity at the minimum attainable p with liftoff at $\cos \approx 0.93$, an occupancy-discrepancy z of -47 ± 4 over 20 seeds, a $3.7\times$ collapse of the per-item field's low tail, exact recovery of the pocket's cardinality and scales by the merge tree (20/20 seeds), and a 26% loss of noise budget ($\sigma^*: 0.203 \to 0.150$). The instrument is calibrated and stress-tested as a whole: liftoff detection uses a global rank-envelope test whose false alarms on 400 pure nulls match nominal rates (a naive pointwise band, by contrast, fired on every null); pocket reporting carries a null-calibrated prominence floor; every readout is stable across reservoir sizes from 1,000 rows; and a detection-power study against a

*The ambit project. Code and reports: `https://github.com/pedapudi/ambit`.

suite of standard diagnostics at matched 1% false alarm shows the continuous layer at full power on pockets as small as 1% of the corpus where mean-cosine, hubness, and kNN-distance baselines sit near the false-alarm floor. We also report negative results, including a measured near-redundancy between the confusion functional and uniformity ($r = 0.978$) whose single order-flip is nonetheless operationally informative. A case study on a public million-document legal corpus and a measurement-driven training recipe with a gradient-locality guarantee complete the report; the recipe is exercised end to end on the case-study corpus and its production encoder, where the held-out measurements reject a misspecified round outright and verify the corrected rounds as safe, with a uniform per-item reduction in held-out cohort collisions (406/408 items, sign-test $p < 10^{-100}$) whose aggregate retrieval effect a label-blind evaluation bounds as positive in point estimate but not significant at 10^3 queries — while the capacity ladder (adapter, LoRA, full fine-tune) shows that the corpus's duplicate pathology is a data problem no training can repair, exactly as the staged protocol routes it. Executing that routing label-blind then delivers the significant gain training could not (confusable-window deduplication: 12 queries fixed, none broken, $p = 5 \times 10^{-4}$, with a measured double dissociation across repair scales), and the same measurements used as *guardrails* on a label-supervised fine-tune halve its collateral multi-hop damage at no cost to its gains — the label-free held-out gate selecting the better model before any judgment is consulted. ambit is unsupervised by design — no labels, gold pairs, or ground truth — and its core depends only on numpy.

1 Introduction

An embedding model maps items to vectors so that semantic relatedness becomes geometric proximity. Once a corpus is embedded — typically onto the unit sphere $\mathbb{S}^{d-1}$, with similarity the cosine of the angle between items — every downstream capability reduces to one primitive: *given a direction, list the items near it.* The primitive degrades in a specific way when distinct items land too close together: queries cannot separate them, and the system exhibits impostors at the top of rankings, near-duplicates that cannot be ordered, and clusters that blur into one another. We refer to this failure mode as *crowding* and to its complement — the corpus's ability to keep distinct items distinct — as *resolution.*

Crowding is a difficult measurement target for two reasons, developed gradually in §2: it is *local* (a collapsing pocket barely moves any global average), and the standard instruments for localizing density — histograms, hexbin lattices, voxel grids, fixed-k neighbor counts — carry hidden discretization parameters whose values change the answer (§5). This report describes ambit, an instrument built to measure occupancy and crowding without either failure. Its contributions:

1. **A first-principles account of why crowding hides** (§2): items as directions, retrieval neighborhoods as spherical caps, the concentration of measure that empties the null's tail, and the arithmetic by which a pathological pocket vanishes into a mean.

2. **A quantitative critique of binned occupancy** (§5), with controlled experiments on the tool's own binned displays: headline lattice statistics are substantially properties of the lattice, not the data.

3. **A continuous, null-calibrated measurement layer** (§6): the exact pair-closeness curve against a uniform and an anisotropy-conditioned null; a one-number occupancy discrepancy via Stolarsky's invariance principle; a per-item distance-to-measure field with a stability guarantee; and a threshold-free merge tree. Every component is exercised on a single traced benchmark — a planted 5% near-duplicate pocket — so the layers can be read against one another.

4. **Operational semantics** (§7): an exact Gaussian confusion law, verified to Monte-Carlo precision; the resolution bandwidth σ^*; a Chernoff identity giving the Wang–Isola uniformity functional operational units; a precise scope statement; and two negative results recorded with the reasons for their failure.

5. **An open instrument and a training recipe** (§9, §10): a streaming numpy-only measurement core, a public million-document case study, and measurement-driven regularizers with a gradient-locality guarantee. (The toolkit delivers its measurements through a self-contained HTML report; that artifact is engineering, not a contribution, and is documented separately in Appendix B.)

6. **A demonstration that the measurements drive interventions** (§10.3–§10.6): a measurement-gated fine-tuning loop on the case-study encoder whose held-out readouts reject a misspecified round that training loss endorsed; a label-blind execution of the instrument's repair prescription whose two scales produce opposite, significant outcomes exactly as prescribed (a double dissociation); and measurement-derived guardrails that

halve the collateral damage of supervised fine-tuning at no cost to its gains, with the label-free gate selecting the better model before any judgment is consulted.

7. **External validation with a measured scope law** (§11): across 28 public corpus–encoder cells scored blind against relevance judgments, the per-item collision field predicts failing documents in every informative cell (22/22, sign test $p \approx 2 \times 10^{-7}$), while the corpus-level σ^* ranks encoders only where relevance is geometric matching and inverts on knowledge-limited tasks — an empirical statement of the alignment–uniformity decomposition that fixes the instrument's scope: per-item triage and comparative geometric budgeting, not encoder leaderboards.

Throughout, ambit is *unsupervised by design*: no labels, gold pairs, or relevance judgments are consumed. We are explicit about the nature of the contribution: the mathematics assembled here is, with small exceptions, classical. The contribution is an *instrument* — the selection, calibration, and composition of known functionals into a measurement system with stated guarantees, stated nulls, and stated scope — together with the negative results that pruned its design space.

2 The Geometry of Crowding, From First Principles

This section develops the geometric considerations that the remainder of the report formalizes. Three elementary observations, taken together, determine the measurement design.

2.1 Items are directions; retrieval neighborhoods are caps

Cosine-compared embeddings live on the unit sphere: each item is a *direction*, and similarity is the angle between directions. A retrieval neighborhood — "everything scoring above 0.8 against this query" — is a *spherical cap*: the disc of directions within a fixed angle of the query (Figure 1). This picture reduces every occupancy question to a statement about caps. Corpus concentration corresponds to caps holding more items than their share of the sphere's area; an item's degree of crowding, to the smallest cap around it that already contains other items; and retrieval confusion, to items sharing a small cap — *future collisions*, since a query landing near one cannot avoid the others.

2.2 In high dimension, "random" means "orthogonal"

Fix one item on $\mathbb{S}^{d-1}$. Almost all of a high-dimensional sphere's area lies in a thin band 90° from it, so a random direction is almost surely near-orthogonal to it: the pair cosine of independent uniform directions concentrates as $\mathcal{N}(0, 1/d)$. At $d = 768$ the standard deviation is $1/\sqrt{768} = 0.0361$; our measured value on 4,000 sampled uniform points is 0.0361 (Figure 2). Three consequences follow. First, "similar" is a tiny window: $\cos = 0.3$ is already an 8σ event for unrelated items. Second, the well-spread null's small-distance tail is *empty* — in 19 of 19 simulated uniform corpora of this size, the fraction of pairs with $\cos \geq 0.30$ is exactly zero — so *any* observed mass there is unambiguous structure. Distance concentration, the classical curse of high dimension [38], is for a null-calibrated instrument a blessing: every close pair is a finding. Third, because almost every pair is a near-orthogonal background pair, *averages*

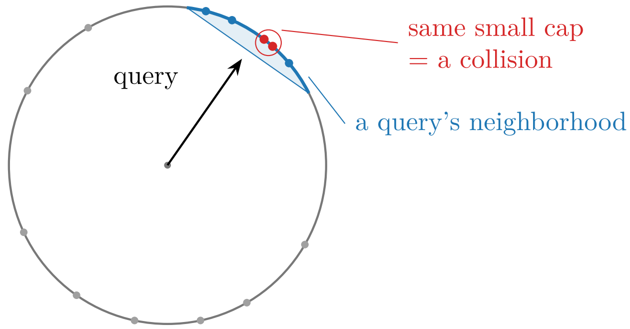

Figure 1: Intuition schematic: items are directions on the unit sphere; a retrieval neighborhood is a cap. Items inside one small cap (red pair) are interchangeable to any query landing there — the geometric definition of a retrieval collision.

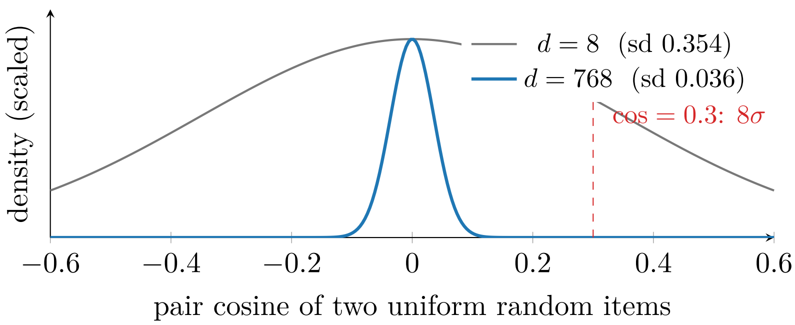

Figure 2: Concentration of measure (analytic densities; the $d = 768$ width is the *measured* sd 0.0361 from 4,000 sampled uniform points, matching $1/\sqrt{d}$ to four decimals): the null pair-cosine distribution collapses to a needle at 0 as dimension grows. The informative region of the null is empty — and the average is dominated by the needle.

over pairs are dominated by the background — the observation the next subsection makes precise.

2.3 Why averages cannot see crowding

Consider the benchmark corpus used throughout this report: $n = 4{,}000$ items at $d = 768$, uniform except for a planted pocket of $k = 200$ near-duplicates (median intra-pocket cosine 0.929) — five percent of the corpus rendered mutually interchangeable, a catastrophic defect for any retrieval system built on it. The pocket contributes $\binom{200}{2}/\binom{4000}{2} \approx 2.5\times10^{-3}$ of all pairs. The mean pairwise cosine therefore moves by about $2.5\times10^{-3} \times 0.93 \approx +0.002$: from 0.000 to 0.002, a shift of 6% of one null standard deviation *per pair* — invisible without a calibrated test, and utterly uninformative about *which* items failed (Figure 3). Covariance-spectrum summaries (effective rank, isotropy scores) are blind for the same arithmetic reason: they are sums over the corpus, dominated by the diffuse majority. More generally: *global averages address whether the whole cloud is degenerate; crowding concerns which part of the cloud has collapsed, at what scale, and which items it contains.* A sound instrument must therefore (i) read *distributions*, not means; (ii) read them *at every scale*, since pockets live at

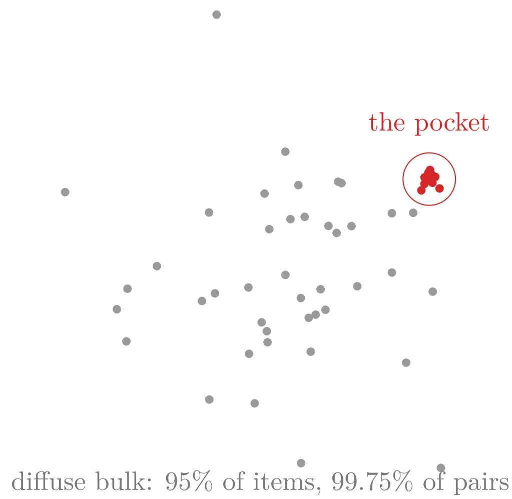

Figure 3: Crowding hides from averages (schematic; annotations from the d=768 benchmark). The planted pocket is catastrophic — 200 items with median mutual cosine 0.93 — yet it owns only 0.25% of pairs, so the corpus mean pair cosine moves from 0.000 to 0.002. No global number locates it; several barely register it.

small scales the bulk never visits; (iii) *calibrate* against a matched null, since absolute cosine is not self-interpreting; and (iv) descend to *named items*, since the actionable output is a list of ids. These four desiderata are the design of §6.

3 Related Work

Anisotropy and resolution. That learned representations occupy a narrow cone was established by Gao et al. [1] and Ethayarajh [2]; Mu et al. [3] showed that removing the mean and a few dominant directions restores much of the usable geometry, and Cai et al. [4] exhibited local manifold structure inside the cone. Global isotropy summaries include effective rank [5] and IsoScore [6]. Wang and Isola [7] decompose contrastive representation quality into alignment and uniformity; the uniformity functional recurs in §7 with new units. Hubness is a high-dimensional pathology of the metric itself [8].

Embedding-quality scores. A family of label-free scores summarizes the covariance or Gram spectrum: RankMe [9], α-ReQ [10], and relatives (surveyed in [11]). A structural fact governs the family: with a linear kernel on unit vectors the nonzero Gram spectrum equals the uncentered second-moment spectrum, so such scores re-parameterize one another; genuinely new signal requires a nonlinear kernel — the Vendi score [12, 13], i.e. effective mode counts from kernel-spectrum entropy [14, 15] — or a different object altogether. ambit adopts this dichotomy as a design filter (§6.5).

Spatial statistics. Ripley's K function [16, 17] reads clustering as a continuous function of scale from the interpoint-distance distribution; on the sphere the null is a cap-area law and, the domain being boundaryless, planar edge corrections vanish [18]; global envelopes provide valid whole-curve inference [19]. The modifiable areal unit problem [22, 23, 24] and the ecological fallacy [25] are the canonical statements of why partition-based statistics are fragile and non-attributive; histogram inefficiency and the origin problem are classical [26, 27, 29], with the averaged shifted histogram [28] the placement-free repair. Stolarsky's invariance principle [20, 21] identifies the mean pairwise chord with the all-caps-all-scales L_2 discrepancy.

Geometric inference. The distance to a measure [30, 31] replaces kernel density estimates and kth-neighbor distances with a quantile-averaged distance function, uniformly Lipschitz in the Wasserstein metric of the input. Mode structure without thresholds is superlevel-set (H_0) persistence: ToMATo [32] and HDBSCAN's condensed tree with excess-of-mass selection [33] are two normalizations of the same object [34], resting on Hartigan's cluster tree [35, 36] with stability from persistence theory [37].

Projections and density. Neighborhood-preserving layouts do not preserve density [41, 42], and ambient kernel density estimation converges at $O(n^{-4/(4+d)})$ — vacuous at d in the hundreds [29]. Local intrinsic dimension estimators [39, 40] are used only per-item and comparatively.

4 Experimental Methodology

Every claim marked *measured* in this report follows one protocol, stated here once. The test environments are uninteresting and are described by their properties in Appendix C; what matters scientifically is the data.

Datasets. Two complementary data regimes carry the evidence, chosen so that between them every claim is testable.

(i) The synthetic benchmark family — known ground truth by construction. Detection, localization, and recovery claims can only be scored against a pathology whose membership and scales are known exactly; planting one supplies that ground truth as *design knowledge* without the instrument ever consuming labels. The traced benchmark of §2.3 is: $n = 4{,}000$ points at $d = 768$ (a contemporary encoder width), drawn as normalized standard Gaussians (exactly uniform on $\mathbb{S}^{d-1}$); the pocket replaces the last $k = 200$ points (5%) with $x_i = (c + s\, g_i)/\|c + s\, g_i\|$ for one shared uniform direction c, i.i.d. Gaussian g_i, and spread $s = 0.01$ — chosen because it lands the pocket in the *near-duplicate regime* (median intra-pocket cosine 0.929, i.e. items a retrieval system cannot rank apart) while leaving every global average essentially unmoved (§2.3): a stringent test for a local diagnostic: the pathology is severe, yet no global statistic responds. The uniform control is the same draw without the replacement. The wider family varies what the traced instance fixes: pocket count (1–20), size (20–800), and tightness ($s = 0.005$–0.06, spanning near-duplicate to loose association); mean-shift cones; low-rank collapses; and pure nulls — the grid behind the sensitivity, calibration, and power studies. All generators are stated in full above or beside their experiments; nothing is tuned to the instrument.

(ii) A public real corpus — realistic geometry at scale. Synthetic corpora cannot exhibit the learned-embedding pathologies that motivate the instrument, so the case study (§9) uses

Nemotron-Pretraining-Legal-v1 [54], a public (CC-BY-4.0) legal corpus published by NVIDIA: 10^6 documents drawn from four of its thirteen subsets — the California Code of Regulations, case-law summaries, CaseHOLD, and eCFR question–answer pairs — embedded at $d = 1024$ by an instruction-tuned text encoder, L2-normalized, stored as sharded Parquet with stable document ids. It supplies what the benchmark cannot: a genuine anisotropy cone (mean pair cosine +0.236), *diffuse* within-subset crowding rather than planted pockets, subset labels used only to *condition* one readout (never consumed by a measurement), and corpus scale that exercises the streaming path. The two regimes are deliberately complementary: the benchmark makes claims falsifiable, the corpus makes them representative.

Standard settings. Unless an experiment sweeps them: pair statistics use 2×10^5 sampled distinct pairs; whole-curve inference uses the global rank-envelope test at 199 null curves (99 inside report figures); the anisotropy-conditioned reference is sampled from the corpus's own covariance eigenvalues at matched pair count; the DTM field uses mass fraction $m = 0.02$; the merge tree uses minimum pocket size 8; the discrepancy z uses 64 null replicates. Each of these defaults is itself validated by a sweep reported in the section that uses it.

Per-experiment protocols. *Lattice fragility* (Table 1): 4,000 synthetic 2-D points (three Gaussian clumps plus uniform background), assigned by the tool's own hexbin code; perturbations are lattice-origin shifts in units of the cell, lattice column counts, and an isotropic data jitter of 0.1% of the data span. *Flip law* (Table 3): for each (θ, σ) cell, two fixed unit vectors at angle θ in $d = 768$; 3×10^5 independent noise draws; empirical flip rate versus the closed form. *Redundancy* (Table 4): eight corpus generators at $n = 2{,}000$, $d = 256$ as named in the table; C and U computed from 6×10^4 sampled pairs each; orderings compared over all $\binom{8}{2}$ pairs. *Training* (Table 8): a planted 300-of-4,000 pocket at $d = 64$; a 20% held-out slice is excluded from mining, weighting, and batching, and *all* reported verdicts (collision counts, neighbor overlap at $k = 10$) are computed on it; schedules as tabulated. *Calibration and power* (§8): 400 pure-null and 50-per-cell pathology corpora at $(n, d) = (2{,}000, 256)$ with 6×10^4 sampled pairs each; every detector is a scalar with a one-sided alarm direction, thresholded at the 1% null quantile computed on the same nulls (the rank test self-calibrates via its p); replicate seeds are disjoint from threshold seeds. *Sensitivity sweeps* (reservoir, pair-sample, mass-fraction, min-size, null-replicate count, noise-model robustness): grids and seed counts as stated inline where each result is used.

Reporting conventions. Aggregates are reported with their null value at matched sample size wherever a null exists; effect sizes are stated in the null's units (z-scores) or as ratios. Negative results are reported with the same prominence as positive ones. Headline benchmark numbers are replicated over 20 seeds and reported as mean ± sd; calibration and power studies use the replicate counts stated with them (400 nulls, 50 per pathology cell, 199-null rank envelopes, 64-replicate discrepancy nulls); single-seed values appear only for deterministic or concentration-driven demonstrations and are marked by the absence of a spread. *Figures* follow one rule: every data-bearing plot embeds unmodified experimental outputs as its coordinates — and the primary examples are the *real corpus's*: the crowding curve, the DTM field, and the collision curves are Nemotron-Legal measurements, and two figures reproduce the instrument's own report readouts on that corpus natively (the named-document localiza-

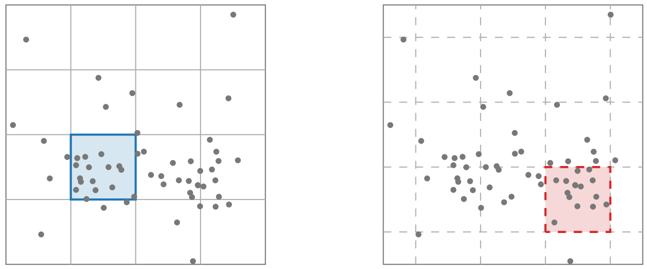

Figure 4: The modifiable areal unit problem on fixed data (the diagram's points and highlighted cells are computed, not sketched): shifting the same lattice by half a cell changes the peak count *and relocates which cell is the peak.* Table 1 gives the effect at scale on the tool's own hexbin display.

Table 1: Lattice fragility on fixed data (n=4,000 synthetic 2-D points; the tool's own hexbin assignment). Perturbing only the *instrument* moves the headline statistics by 30% (placement) and 2.4× (scale); a 0.1%-of-span data jitter, by contrast, moves the peak by one count — fragility is to the lattice, not the data, until a point crosses a boundary.

perturbation	peak-cell count	occupied cells
lattice as placed	310	256
lattice shifted 0.25 cell	310	253
lattice shifted 0.5 cell	403	262
20 columns	574	168
26 columns	340	256
34 columns	242	350
data jittered by 0.1% of span	339	—

tion list and the pockets bar, with the report's values verbatim); the lattice demonstration's highlighted peak cells are computed from the figure's own points; the remaining figures are intuition schematics and are captioned as such.

5 The Fragility of Binned Occupancy

ambit's early density views were conventional: hexbin lattices over a 2-D projection, voxel grids, radial shells, fixed-k neighbor counts. This section records, quantitatively, why the measurement layer abandoned them.

5.1 The answer belongs to the lattice

Figure 4 and Table 1 show the two MAUP effects [24] on the tool's own display: *zonation* (placement: a half-cell shift moves the headline peak count from 310 to 403 and relocates the peak cell) and *scale* (a factor 2.4 across reasonable column counts). Neither perturbation touches the data. The classical demonstration that re-zoning fixed data can move a correlation across essentially $[-1, +1]$ [23] is the extreme form of the same phenomenon. Histograms additionally pay an efficiency tax for discretization — MISE $O(n^{-2/3})$ against $O(n^{-4/5})$ for kernel estimators [26, 27] — and averaging over placements (the ASH [28]) recovers both placement-independence and the kernel rate. ambit retains one hexbin *display*, de-fragilized exactly this way, captioned as a view, not a measurement.

5.2 One hidden scale each; duplicate blindness

Crowding is multi-scale: near-duplicate pockets live at tiny angles, topical clumps at moderate ones, the anisotropy cone at large ones. Every bin width, shell count, and fixed neighbor count k silently commits to one scale. Injecting 200 near-identical points inside an already-dense cell moves the hexbin peak from 340 to 540 — indistinguishable from a benign hot cell — while the continuous per-item field of §6.3 collapses by a factor of 18 at its first percentile. A fixed k in a kNN graph is the same defect in rank-based form: in a dense pocket k=10 spans a tiny radius, in a sparse rind a huge one, so "neighborhood" denotes a different physical scale at every point — precisely what a crowding measure must not let drift.

5.3 Steps, nulls, names, and projections

Four further costs are structural. *Instability:* a bin count is a step function of the data with no perturbation guarantee; the field of §6.3 is provably Lipschitz in the Wasserstein distance of the input and measured stable (self-correlation 0.997 under jitter at d=768). *No null:* a peak count carries no statement of what a well-spread corpus would show — a perfectly uniform cloud still has a hottest cell. *No names* (the ecological fallacy [25]): a cell count cannot say which items are crowded; the actionable output of a diagnosis is a list of ids. *Projection:* binning a 2-D layout measures the shadow — far-apart groups collide in a cell, orthogonal crowding vanishes, and nonlinear layouts actively destroy density [41]. The repair is not a better ambient density estimate (kernel density estimation is hopeless at d in the hundreds [29]) but the abandonment of partitions for scale-free functionals of interpoint distances, whose guarantees carry no dimension factor.

6 Continuous, Null-Calibrated Occupancy

Let $x_1, \ldots, x_n \in \mathbb{S}^{d-1}$ (unit-normalized; cosine $c_{ij} = \langle x_i, x_j \rangle$; chord $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$). ambit computes exact full-corpus second moments by streaming and draws a bounded uniform reservoir plus a large random-pair cosine sample (2×10^5 pairs by default) for everything else. Throughout this section the *figures* show the real corpus (Nemotron-Legal, §4) — the primary examples are drawn from learned-embedding geometry rather than synthetic constructions — while the traced benchmark of §2.3, whose planted pathology is known exactly, supplies the falsifiable counterpart for every layer in the text.

6.1 The crowding curve

Definition 1 (Pair-closeness curve)**.** $K(t) = \Pr_{(i,j)}[\, c_{ij} \geq t\,]$*: the fraction of distinct pairs at least t-similar, estimated* exactly *from the pair sample by sorting — no bins, no origin, no* *width.*

K is Ripley's K function on the sphere [16] in cosine coordinates. Two references with matched sample size accompany it (Figure 5):

(i) The uniform null, tested globally. For i.i.d. uniform points the pair cosine satisfies $c = 2B - 1$, $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$, so null pair samples are drawn directly — no d-dimensional vectors are materialized. The sphere being boundaryless, no edge correction exists to get wrong [18]. Inference over the whole curve is a one-sided global *rank-envelope* test [19] with extreme-rank-length tie breaking (a one-sided K test is massively tied where every null is exactly zero): the data curve is ranked among 199 null curves, a single global p covers all scales at once, and *liftoff* — the highest cosine at which the data exceed the α-critical envelope — is annotated only when the global test rejects. Calibration is measured, not assumed: on 40 pure-null corpora the gated test reported 0 liftoffs, whereas the naive pointwise band (the max of the null curves) fired on *every* null seed at the bulk edge — a direct demonstration of the multiple-comparison failure — and even the α-critical envelope alone over-fires $\approx 4\times$ nominal; the envelope locates the scale, the rank p decides significance.

(ii) The anisotropy-conditioned null. Real corpora are cone-shaped for benign reasons; judging them against uniformity alone over-alarms. The second reference is the *angular central Gaussian* with the corpus's own covariance spectrum — $x = \Sigma^{1/2}g/\|\Sigma^{1/2}g\|$, sampled from the eigenvalues alone by rotation invariance: the corpus's second-moment cone *without* its clustering. Height of the data curve above this reference is crowding the cone cannot explain.

Figure 5 shows the curve on the real corpus (Nemotron-Legal, §4): the global test rejects at the minimum attainable $p = 1/200$ with liftoff at $\cos \approx 0.81$. The reading is more detailed than a single scalar: through mid-similarities the data track the ACG reference — that part of the excess over uniformity *is* the anisotropy cone, benign and expected of a learned encoder — while above $\cos \approx 0.8$ the data rise above both references: genuine tight-pair structure at the near-duplicate scale, which §6.3 will attribute to named documents. On the traced benchmark the same test rejects at $p = 1/200$ with liftoff $\cos \approx 0.93$ and a shoulder of excess mass exactly matching the planted pocket's pair share (2.4×10^{-3}) — the counterpart with known ground truth.

6.2 The occupancy discrepancy (Stolarsky z)

Proposition 1 (Stolarsky [20])**.** *For points on $\mathbb{S}^{d-1}$, the spherical-cap L_2 discrepancy — the squared deviation between empirical and expected cap occupancy, integrated over* all *cap centers and* all *cap radii — equals, up to explicit constants, a constant minus the mean pairwise chord $\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$.*

The mean pairwise chord is therefore "every lattice at every scale at once": the exact completion of what binned occupancy approximates, with no lattice to choose. `ambit` reports it as a z-score against the uniform null at matched pair-sample size, pricing in sampling noise. Measured on the benchmark over 20 seeds: the uniform control scores $z = -0.14 \pm 1.00$ — exactly calibrated — while planting the 5% pocket moves the mean chord from 1.41405 to 1.41160, a shift of 0.17%, which the concentrated null resolves as $z = -47 \pm 4$. Tightening the pocket further (members within a cap of angular radius ≈ 0.15) drives the same statistic

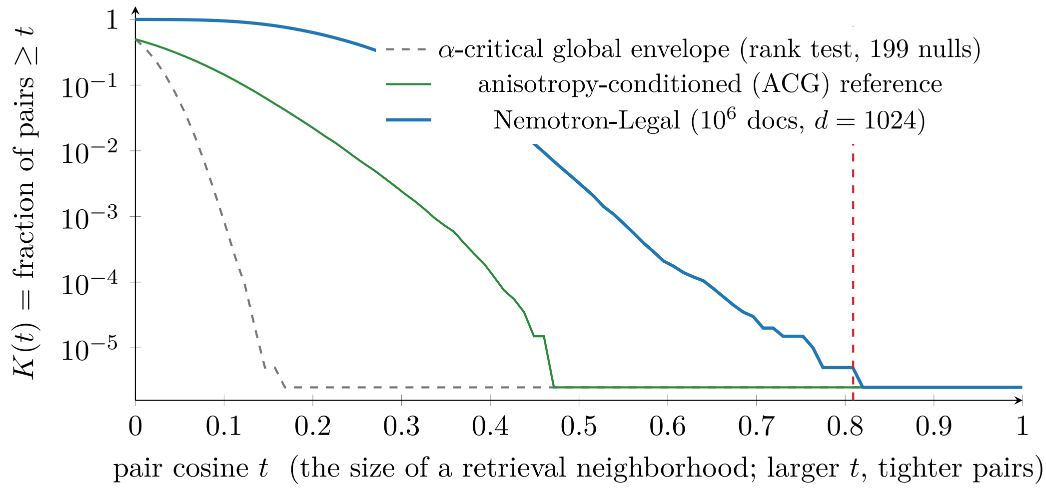

Figure 5: The crowding curve on the real corpus (Nemotron-Legal [54], measured from the report's own $2{\times}10^5$-pair sample). Through mid-similarities the data track the anisotropy-matched reference — that excess over uniformity is the cone, benign — but above cos ≈ 0.8 the curve rises above both references: genuine tight-pair structure, globally significant at the minimum attainable $p = 1/200$. No bin width or lattice placement exists anywhere in the figure, and the liftoff claim is family-wise valid across all scales.

to $z = -430$: detection strength scales with the pathology while the control stays at noise level.

Two properties of z are stated so it is read correctly. First, *z is a test statistic, not an effect size*: its magnitude grows with the pair-sample size P (the null mean's spread shrinks as $1/\sqrt{P}$), so z values are comparable only at matched P and the instrument reports them at its standard $P = 2{\times}10^5$; σ^* and the liftoff scale, by contrast, are effect-size-like and concentrate as $1/\sqrt{P}$ (measured: sd(σ^*) falls $0.0126 \to 0.0013$ over $P = 10^4 \to 10^6$). Second, the null's spread must itself be estimated well: with 24 null replicates $|z|$ is inflated $\approx 20\%$ and reproduces with sd ≈ 8; the default is 64 replicates ($z = -43 \pm 4.5$ on the benchmark), measured across a replicate sweep.

6.3 Localization: the distance-to-measure field

Definition 2 (Empirical DTM [30, 31])**.** *For mass fraction m and $k = \lceil mn \rceil$:* $\mathrm{DTM}_m(x_i) = \left(\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2\right)^{1/2}$*, with $r_{i,(j)}$ the distance to x_i's jth nearest reservoir point — the radius the item needs to gather an m-share of the corpus.*

Small radius means crowded; large, isolated (Figure 6). The one knob m is a *mass fraction* — scale-free in n — and replaces bin width, grid origin, and fixed k simultaneously. Its one trade-off is measured rather than asserted (Table 2): separation between a planted pocket and the bulk collapses almost exactly at the predicted blur boundary $m \approx k/n$, so the reporting rule is "choose m below the smallest pocket share you care about," and the default $m = 0.02$ resolves pockets down to 2% of the corpus.

The estimator inherits the stability theorem $\|\mathrm{DTM}_m^P - \mathrm{DTM}_m^Q\|_\infty \leq m^{-1/2}\, W_2(P, Q)$ [30]:

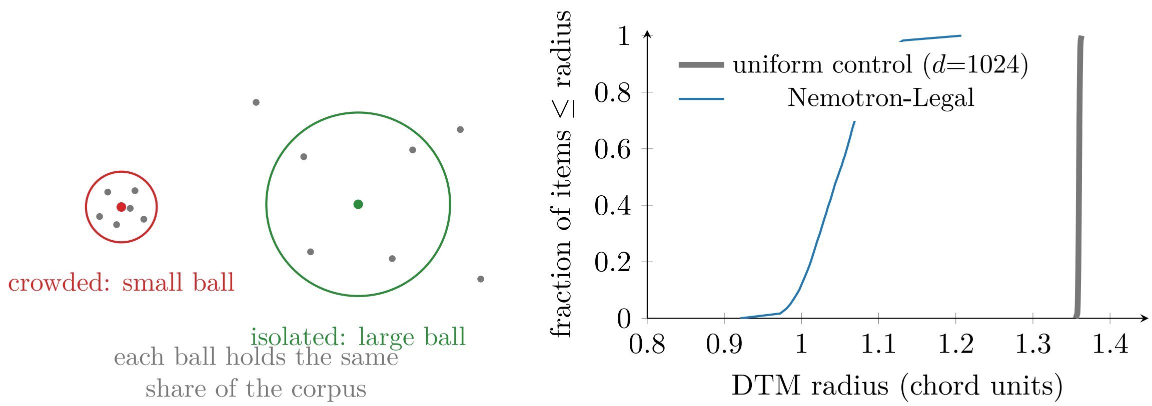

Figure 6: The distance-to-measure field. *Left:* the intuition schematic — each item is scored by the radius of the ball it needs to gather a fixed share ($m = 2\%$) of the corpus. *Right:* the field's exact CDF on the real corpus (measured): the *entire* field sits left of the uniform needle at 1.36 — the anisotropy cone shortens every radius, so the uniform band is diagnostic of the cone, not of pockets — and the finding is the low tail against the corpus's own bulk (1st percentile 0.966 against median 1.047), whose members the instrument names by id (Figure 7).

Table 2: The DTM mass fraction's measured blur boundary (bulk median / pocket median of the field; benchmark family, planted pocket of k members in $n = 4{,}000$). Separation survives while $m \lesssim k/n$ (boldface) and collapses beyond it — exactly the predicted rule.

	m=0.005	m=0.01	m=0.02	m=0.05	m=0.1
$k = 20$ (k/n=0.005)	**2.83**	1.34	1.13	1.05	1.02
$k = 50$ (k/n=0.013)	**3.63**	**3.59**	1.52	1.14	1.06
$k = 200$ (k/n=0.050)	**3.70**	**3.69**	**3.67**	**3.52**	1.37
$k = 800$ (k/n=0.200)	**3.75**	**3.74**	**3.74**	**3.72**	**3.70**

a small perturbation of the corpus provably moves every item's score a small amount — the guarantee step functions lack, and measured: jittering every coordinate by 10^{-3} leaves the field 0.997-correlated with itself, and against a fixed 10^6-row corpus the field's first percentile is invariant to the reservoir drawn (0.365–0.366 across reservoirs of 10^3–1.6×10^4 rows, ten seeds each), as is every other headline readout — σ^* spans 0.0559–0.0562 against 0.0564 at a 3.2×10^4 reference. On the traced benchmark the planted pocket drags the low tail to 0.364 against a bulk at 1.35 — the 3.7× separation behind the detection-power results of §8.

Both tails are actionable and *named*: the low tail is the crowded-item list (each entry annotated with its expected collision count at the corpus's own σ^*; §7), the high tail the coverage voids. In strongly cone-shaped corpora the entire field sits below the uniform band — the cone shortens every radius — and the instrument reports the regime explicitly rather than an uninformative all-items count; the finding in that regime is the shape of the low tail against the corpus's own bulk.

most crowded documents

id · DTM radius · ≈collisions at σ^*=0.12		
`9b3dd305-0e50-4…`	0.921	≈12
`a310931d-78cc-4…`	0.922	≈12
`369ba45c-63b5-4…`	0.923	≈4
`7ce46beb-eb01-4…`	0.928	≈4
`18a8b3f6-a37c-4…`	0.933	≈3
`efaeae26-62ef-4…`	0.936	≈4

most isolated (voids)

`59c2a211-f8cc-4…`	1.207
`2a761d0e-c3e0-4…`	1.198
`0eba585f-60f9-4…`	1.190

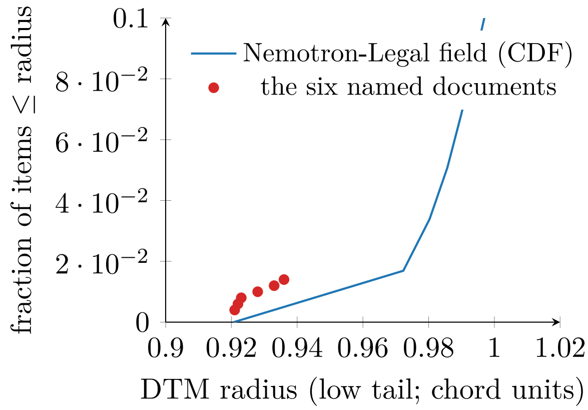

Figure 7: The per-entity localization readout, reproduced natively from the instrument's report on Nemotron-Legal (values are the report's own). *Left:* the crowded-document list — *named by uuid*, each with its DTM radius and its expected collision count at the corpus's σ^*: the radius says how crowded, the count says what it costs. *Right:* where those six sit — the extreme low tail of the field's CDF (the bulk rises to 1 near radius 1.05; the uniform band lies far right at 1.36). Aggregate, cell-based displays cannot provide this output.

6.4 Structure: the condensed merge tree

Between the global curve and the per-item field sits *structure*: one broad pile, or many tight pockets, and who constitutes each? `ambit` surfaces the hierarchy its clustering backend already computes and previously flattened: single linkage on the mutual-reachability metric [33], condensed at a minimum pocket size, selected by excess of mass in $\lambda = 1/\text{distance}$ — equivalently H_0 persistence with the elder rule [32, 34]. Each selected pocket reports birth, death, prominence, size, and member ids (Figure 8). Three implementation findings mattered — the third found by the validation battery itself. First, the naive elder rule is wrong for this purpose: the tightest pocket is the *eldest* component and survives to the root, so the bulk "dies into it"; excess-of-mass selection resolves the inversion. Second, single-linkage chains inside a tight pocket are caterpillars, so membership is reported as the condensed core plus side branches that fell out in the tight regime (below the geometric mean of birth and death), excluding bulk stragglers. Third, chaining can prevent *any* real split from existing: a tight pocket may seed the one component that absorbs the bulk point by point, so no node ever has two $\geq$ min-size children, the only cluster is the root — and the tightest structure in the corpus goes silently unreported (a parameter sweep exposed this: at min-size 32 a planted 200-pocket returned *nothing*, and 3 of 20 benchmark seeds missed it even at the default). The fix treats a childless root as a candidate with death at the final merge scale, whereupon the membership rule extracts exactly the pocket core. Post-fix, the tree recovers the planted pocket in 20 of 20 seeds at exact size (200.0 ± 0.0; birth 0.063, death 0.898, prominence 0.835) across min-sizes 8–64, and on a two-pocket variant recovers sizes 60 and 40 exactly. Reported prominence also carries a *null floor*: uniform corpora of matched size and dimension produce maximum prominences of only ≈ 0.005 (99th percentile 0.009 over 400 nulls), so the instrument simulates the floor per run and suppresses pockets within it — a genuine pocket sits two orders of magnitude above it. Figure 8 shows the resulting readout

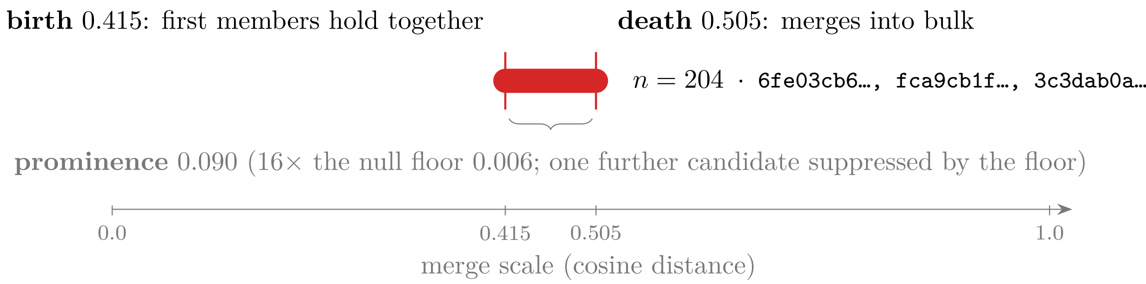

Figure 8: The pockets readout, reproduced natively from the instrument's report on Nemotron-Legal with the merge-tree anatomy annotated on it: the one pocket surviving the null-calibrated prominence floor (204 members, sample ids on the bar) is *born* when its first members hold together and *dies* into the bulk (excess-of-mass selection). Its prominence 0.090 is modest — a loose association, not a duplicate cluster — yet 16× the null floor, and therefore reportable; and the floor suppressed one further candidate; on the traced benchmark the same construction recovers a planted pocket exactly (200/200; birth 0.063, death 0.898, prominence 0.835), with no flat threshold anywhere.

on the real corpus: one pocket survives the floor (204 members, sample ids on its bar), and one further candidate was suppressed.

6.5 Complexity, and the design filters

All continuous-layer statistics run on the reservoir or the pair sample: the curve and its nulls are $O(P \log P)$ in the pair-sample size; the DTM field is a blocked Gram top-k, $O(n_r^2 d)$ on a capped subsample (seconds on CPU); the merge tree is a dense Prim MST on $n_r \leq 4{,}096$ rows plus a linear condensation pass. Full-corpus second moments stream in one pass.

Candidate metrics were admitted by five filters, recorded because they prune future proposals cheaply: *spectrum* (does it reduce, under a linear kernel, to the covariance spectrum already reported?), *continuity* (a continuous functional, or a step function of bin/rank/partition membership?), *null* (does it come with a calibrated reference?), *naming* (can it score items, or only regions?), and *supervision* (label-free?). These excluded, among others, RankMe, NESum, and stable rank (spectrum), k-means-conditioned statistics (continuity), raw counts (null), all cell aggregates (naming), and neural-collapse metrics [44] (supervision).

7 Operational Semantics: the Resolution Bandwidth

Geometric functionals quantify how uniformly a corpus covers the sphere; a practitioner needs to know how much query perturbation the corpus can absorb before retrieval degrades. This section converts the former quantity into the latter exactly, then measures the conversion.

7.1 An exact pairwise confusion law

Model a query aimed at item x as $q = x + \sigma g$, $g \sim \mathcal{N}(0, I_d)$ — the minimal model of a query that intends x but lands nearby. Retrieval errs on competitor y when $\langle q, y\rangle > \langle q, x\rangle$.

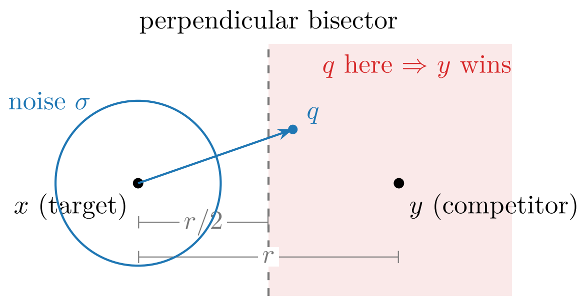

Figure 9: The confusion law's geometry: the decision boundary between target and competitor is the perpendicular bisector at distance $r/2$; Gaussian noise of scale σ crosses it with probability exactly $\Phi(-r/2\sigma)$. A far competitor is safe; a near-duplicate ($r \to 0$) has error probability approaching 1/2 at every noise scale.

Table 3: Verification of Proposition 2 by direct simulation at $d = 768$ (3×10^5 trials per cell): empirical flip rate vs. $\Phi(-r/2\sigma)$ with $r = 2\sin(\theta/2)$. Maximum absolute deviation over the grid: 9×10^{-4}, i.e. Monte-Carlo error — the law is exact, not asymptotic.

	$\sigma = 0.05$		$\sigma = 0.15$		$\sigma = 0.30$		$\sigma = 0.60$	
θ	emp.	pred.	emp.	pred.	emp.	pred.	emp.	pred.
0.1	.1583	.1588	.3700	.3695	.4335	.4338	.4677	.4668
0.3	.0014	.0014	.1592	.1596	.3101	.3092	.4021	.4017
0.6	.0000	.0000	.0245	.0244	.1626	.1623	.3113	.3112
1.0	.0000	.0000	.0007	.0007	.0554	.0550	.2114	.2121
1.4	.0000	.0000	.0000	.0000	.0156	.0159	.1416	.1415
2.0	.0000	.0000	.0000	.0000	.0024	.0025	.0803	.0804

Proposition 2 (Exact flip law)**.** *For unit vectors at chord distance $r = \|x - y\|$,* $\Pr[\langle q, y\rangle > \langle q, x\rangle] = \Phi\big(-r/2\sigma\big)$*, exactly, in any dimension.*

Proof. $\langle q, y - x\rangle$ is Gaussian with mean $\langle x, y - x\rangle = \cos\theta - 1 = -r^2/2$ and standard deviation $\sigma\|y - x\| = \sigma r$; the flip probability is $\Phi\big((-r^2/2)/(\sigma r)\big)$. Geometrically (Figure 9): the boundary is the bisector at distance $r/2$, crossed with probability $\Phi(-(r/2)/\sigma)$. ☐

Table 3 verifies the law across a 6×4 grid of angles and noise scales to a maximum deviation of 9×10^{-4}. Two readings follow: a competitor far relative to the noise is harmless, and a near-duplicate has error probability approaching 1/2 at every noise scale — the formal reason duplicate pockets are the worst crowding pathology (cf. §6.1's liftoff at the near-duplicate scale).

7.2 $C(\sigma)$, σ^*, and the uniformity bound

Summing the law over competitors gives the expected number of items outranking the intended target,

$$C(\sigma) = \sum_{j\neq i} \Phi\Big(-\frac{r_{ij}}{2\sigma}\Big) \;\approx\; (n-1)\int \Phi\Big(-\frac{r}{2\sigma}\Big)\, dK(r),$$

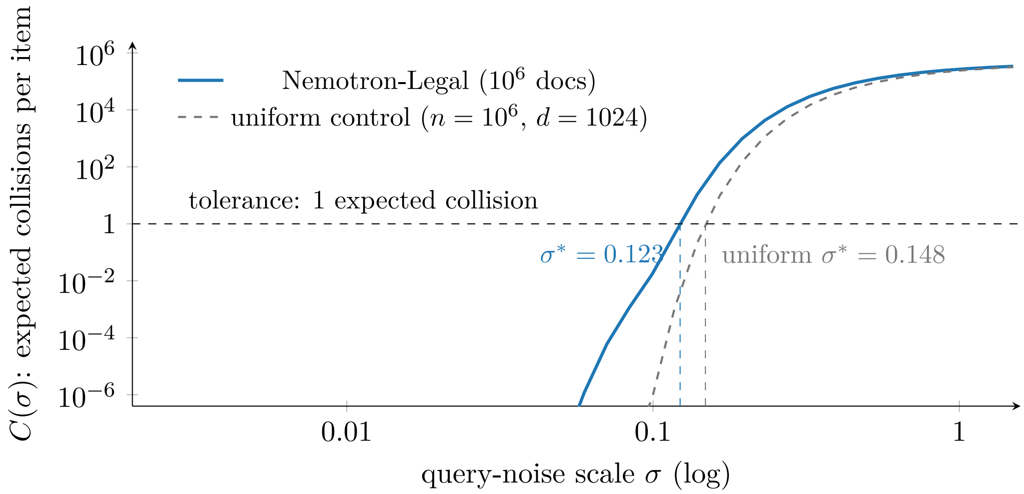

Figure 10: Expected collisions vs. query noise on the real corpus (measured; Nemotron-Legal). The curves separate exactly in the decision region — the corpus's cone and tight pairs dominate $C(\sigma)$ at the noise scales where retrieval operates, pulling the tolerance crossing from $\sigma^* = 0.148$ to 0.123: the corpus retains 83% of a well-spread corpus's noise budget. At large σ the curves converge (every pair confuses) and at tiny σ both vanish; the tolerance crossing is determined where they differ. On the traced benchmark the same construction attributes a cost of 26% of the budget to the planted 5% pocket (σ^*: 0.203 $\to$ 0.150).

a functional of the pair-distance distribution alone. By Boole's inequality the sum bounds the probability of *any* error under arbitrary competitor correlations, so C is conservative: hubness makes it loose, never wrong.

Definition 3 (Resolution bandwidth)**.** *σ^* is the largest σ with $C(\sigma) \leq 1$: the most noise queries can carry before, in expectation, one wrong item outranks the right one. C is monotone; bisection inverts it.*

Figure 10 shows the real corpus's curve against its uniform control: Nemotron-Legal crosses the one-collision tolerance at $\sigma^* = 0.123$ against 0.148 for a well-spread corpus of the same size and dimension — it retains 83% of the ideal noise budget — while on the traced benchmark the planted 5% pocket spends 26% of the budget ($\sigma^* : 0.203 \to 0.150$). The per-coordinate scale σ translates to intuition through the query's self-similarity decay $\langle \bar{q}, x \rangle \approx (1 + \sigma^2 d)^{-1/2}$: σ^* states how *faded* a query may become against its own target and still be expected to win ($\sigma^* = 0.123$ at $d = 1024$ is a query decayed to cosine ≈ 0.25).

Proposition 3 (Uniformity as a Chernoff bound)**.** *With $U_t = \log \mathbb{E}\, e^{-t\|x-y\|^2}$ the uniformity functional of [7]: $C(\sigma) \leq (n-1)\, e^{U_t}$ at $t = 1/(8\sigma^2)$.*

Proof. $\Phi(-z) \leq e^{-z^2/2}$ with $z = r/2\sigma$. □

The widely used uniformity temperature is thus not arbitrary: $t = 2$ is a collision bound at query noise $\sigma = 0.25$. ambit reports uniformity with this reading attached.

Table 4: Measured near-redundancy of the confusion functional with uniformity across eight synthetic corpus types ($n = 2{,}000$, $d = 256$; C at $\sigma = 0.25$, U at the matched $t = 2$). Rank correlation 0.978; 27 of 28 pairwise orderings agree. The one disagreement (boldface) is between qualitatively different pathologies — and the confusion ordering is the operationally correct one.

corpus type	$C(0.25)$	$U_{t=2}$
uniform	4.86	−3.97
one clump of 50	4.90	−3.96
twenty clumps of 40	5.41	−3.90
one clump of 200	5.58	−3.87
five clumps of 100	5.80	−3.84
anisotropic cone	8.38	−3.48
low-rank collapse	**15.49**	**−3.08**
cone + duplicate pocket	**17.78**	**−3.21**

7.3 Scope

The semantics carry their boundary: (i) the corpus is treated as its own query population — exact for deduplication, clustering, and corpus-as-queries retrieval; extrapolation to an external workload assumes the workload lands where the documents are; (ii) the noise channel is isotropic, while real query drift is structured — and the sensitivity of this assumption is *measured*: re-ranking a grid of corpus types under structured Gaussian channels at matched total power, the isotropic ordering survives moderately structured noise (rank-25 subspace noise: 0.99 pairwise-order agreement over five seeds) and degrades under extreme misspecification (rank-2: 0.79; noise aligned with each corpus's own top principal directions: 0.82), so comparative use is robust and adversarial noise structure is the stated failure mode; (iii) the aggregate is an upper bound. σ^* is therefore a conservative intra-corpus guarantee, strongest used comparatively — one corpus embedded two ways, or one embedding over time.

7.4 Negative results

Two negative results from the design process are recorded to prevent their re-derivation.

Near-redundancy of C with uniformity (Table 4). As a corpus-level *discriminative* measurement, the confusion transform adds almost nothing beyond uniformity — rank correlation 0.978, 27/28 order agreements — its value being semantic: units, calibration, per-item counts. The single disagreement is informative: the confusion kernel ranks a duplicate-pocket corpus as worse than a diffuse low-rank collapse, the Gaussian kernel oppositely; duplicate pairs approach error probability 1/2 at every noise scale while a spread subspace remains partially navigable, so the confusion ordering is the operationally correct one, and the flip occurs precisely between qualitatively different pathologies.

A vacuous stability certificate. Composing the DTM stability bound with a split-half empirical Wasserstein estimate, hoping for finite-sample error bars, fails structurally at embedding dimension: the matching distance between independent halves is of the order of the data scale (≈ 1.35 on $\mathbb{S}^{767}$ at $n_r = 2{,}000$), so the resulting certificate is $m^{-1/2}\widehat{W}_2 \approx 13.5$ — an order of magnitude beyond the field's entire range. Dimension-free certificates would need a different metric with its own stability theorem — an open question, not an engineering task.

Table 5: Detection power at matched 1% false alarm (50 replicates per cell; thresholds calibrated on 400 nulls; $n = 2{,}000$, $d = 256$). Pocket cells are (count $\times$ size, spread s; $s = 0.005$ is near-duplicate, 0.06 loose). Continuous-layer detectors left of the rule; baseline suite right. The hardest cells — a single pocket of 1% of the corpus — separate the layers; boldface marks the exceptions discussed in the text.

cell	rank_p	σ^*	DTM	pocket	meancos	Iso	unif.	hub	kNN_{q05}
1×20, s=.005	1.00	1.00	1.00	1.00	.04	**1.00**	.88	.18	.22
1×20, s=.02	1.00	1.00	1.00	1.00	.02	**1.00**	.70	.42	.34
1×20, s=.06	1.00	.22	1.00	1.00	.04	**1.00**	.02	.90	.32
1×50, s=.005	1.00	1.00	1.00	1.00	.58	1.00	1.00	1.00	.92
5×20, s=.02	1.00	1.00	1.00	1.00	.26	1.00	1.00	1.00	1.00
1×200, any s	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
cone (mean shift)	1.00	1.00	1.00	**.12**	1.00	**.00**	1.00	1.00	1.00
low-rank collapse	1.00	1.00	1.00	1.00	.26	1.00	1.00	**.00**	1.00

8 Calibration and Detection Power

The instrument's statistics were calibrated and stress-tested as a whole against 400 pure-null corpora, then compared for detection power against a suite of standard diagnostics with *every* threshold set on the same nulls at a matched 1% false-alarm rate — a comparison at fixed specificity across all detectors.

Null calibration (E4b). On pure nulls ($n = 2{,}000$, $d = 256$; the benchmark family's generators with no pathology): the global rank test rejects at $p \leq 0.01$ in 1.25% of nulls (nominal 1%); the gated liftoff fires in 6% at $\alpha = 0.05$ (nominal 5%); $|z| > 3$ occurs in 0.75%; and the pocket-prominence maximum has null median 0.005 and 99th percentile 0.009 — the floor of §6.4. Measured false-alarm rates for all calibrated thresholds land on 0.010–0.013.

Detection power (E2). Table 5 reports the pathology grid. Three observations follow. First, the continuous layer — the rank test, σ^*, the DTM tail ratio, and pocket prominence — is at *full power in every pocket cell*, including the hardest: a single pocket of 20 items (1% of the corpus), where mean-cosine (0.04), hubness (0.18), and the kNN-distance quantile (0.22) sit near the false-alarm floor — quantifying the claim of §2.3. Second, the exceptions are reported explicitly: covariance-spectrum statistics (IsoScore) are *also* at full power on single coherent pockets — a 1% mass sharing one direction lifts an eigenvalue past the null spectrum edge — so for pure corpus-level detection of one tight pocket, a spectrum summary suffices; what it cannot do is name the members, resist a mean-shift cone (power 0.00 there, since centering removes the shift), or express the loss in operational units. Third, the discriminating cells behave as the theory says they should: the mean-chord z is weak on small pockets (their pair mass is tiny); the pocket detector correctly declines to alarm on a cone (0.12 — a cone is not a pocket); and hubness is blind to a low-rank collapse. The layer's value is therefore not a single more powerful scalar but full power at matched specificity *together with* localization to named items, scale attribution, and operational units — the properties the baselines structurally lack.

Table 6: Scaling on the measurement host (numpy-only core; raw `.npy` input; per-phase wall time and peak RSS from forked children — RSS is dominated by memory-mapped input pages, not allocator footprint). Left: corpus-size sweep at $d = 1024$; the streaming scan is linear in bytes at $\approx 2\,\mathrm{GB/s}$ and the reservoir-bound phases are *constant* in n. Right: dimension sweep at $n = 10^6$; the scan carries the d^2 scatter accumulation and the context build the d^3 eigendecomposition, while the continuous layer stays nearly flat.

	$d = 1024$				$n = 10^6$		
n	scan	ctx	cont.	d	scan	ctx	cont.
10^5	0.2 s	2.7 s	1.8 s	256	0.6 s	0.5 s	1.8 s
10^6	2.0 s	3.0 s	1.8 s	1024	2.0 s	3.2 s	1.8 s
$3{\times}10^6$	5.6 s	3.0 s	1.8 s	2048	4.6 s	10.4 s	2.0 s
10^7	18.6 s	2.6 s	1.8 s	4096	12.4 s	52.6 s	3.4 s

9 The Instrument

9.1 Architecture

`ambit` is a library with a thin CLI. One canonical in-memory type (`EmbeddingSet`) flows through a pipeline: a *streaming scan* (one pass over `.npy/.npz/.parquet/.csv/.jsonl`, single file or sharded directory) accumulates exact second moments, norms, and a uniform reservoir; a context builder derives projections (PCA by default), the eigenspectrum, a kNN graph (optionally filtered to *reciprocal* neighbors, which suppresses hubs by construction), unsupervised clusters, and the pair-cosine sample. All measurements of §6–§7 are library functions over these objects; a separate presentation layer (Appendix B) renders them. The core depends on `numpy` alone; heavier capabilities (tabular IO, ANN graphs, UMAP, GPU paths, on-the-fly embedding via an OpenAI-compatible endpoint, the training extra) are optional and degrade gracefully. A comparison mode aligns two embeddings of the same items by id and computes a local neighbor-overlap view read against global representational-similarity statistics (CKA [43], kernel and rigid distances); the local/global pair can disagree informatively — a high-CKA, low-overlap outcome is a re-skin that preserves gross structure while rewriting fine neighborhoods.

9.2 Scaling

Table 6 gives the measured curves. The pipeline behaves as designed: the one pass over the corpus is I/O-bound and linear in bytes (a $10^7 \times 1024$, 41 GB corpus streams in 18.6 s on raw `.npy`; the Parquet path adds decode overhead — the case study's $10^6 \times 1024$ sharded-Parquet run takes ≈ 3.5 minutes end to end on the smaller host), while everything downstream of the scan is *reservoir-bound and therefore constant in corpus size*: context build ≈ 3 s and the entire continuous layer ≈ 1.8 s whether the corpus has 10^5 or 10^7 rows. Dimension enters through the scan's d^2 scatter accumulation and the context build's d^3 eigendecomposition (52.6 s at $d = 4096$ — the one super-linear phase), while the continuous layer grows only through BLAS on the reservoir Gram ($1.8 \to 3.4$ s over $d = 128 \to 4096$).

One caveat is demonstrated empirically. With the planted pocket placed in the corpus's *last* rows (the layout produced by label-sorted shards), the full scan detects it (rank test

Table 7: Case study: Nemotron-Pretraining-Legal-v1 [54] (four subsets, 10^6 documents), $d = 1024$; full report in $\sim$3.5 minutes on CPU, one streaming pass.

items × dims	1,000,000 × 1024
mean pair cosine	+0.236
IsoScore / effective rank	0.090 / 728.0/1024
occupancy discrepancy z	−4,211
whole-curve rank test	$p = 0.005$; liftoff cos ≈ 0.82
resolution bandwidth σ^* (uniform null)	0.123 (0.148)
noise budget retained	83%
top pocket (size; birth; prominence)	204; 0.41; 0.09
kNN purity / silhouette (4 provided subsets)	0.95 / +0.04
most-crowded items	$\approx$ 3–12 expected collisions each

$p = 0.010$, liftoff 0.92), while approximate scans of the first $5{\times}10^4$ to $3{\times}10^5$ rows see a clean corpus ($p = 0.18$–0.45, no liftoff, mean cosine ≈ 0): an approximate scan's reservoir is only as representative as the row order that fed it, so the instrument's approximate mode is for exploratory iteration rather than for conclusions on data whose layout may correlate with content.

9.3 Case study: a million-document corpus

Table 7 summarizes a full run on the Nemotron-Pretraining-Legal-v1 corpus [54] (§4): 10^6 documents from four legal subsets, embedded at $d = 1024$ by an instruction-tuned text encoder. The corpus is strongly cone-shaped (mean pair cosine +0.236; IsoScore 0.090) yet spectrally rich (effective rank 728/1024); the occupancy discrepancy sits at $z = -4{,}211$, and the whole-curve rank test rejects uniformity at the minimum attainable $p = 0.005$ with liftoff at cos ≈ 0.82 — genuine tight-pair excess, at the near-duplicate scale. (A correction attributable to the calibration study: an earlier pointwise reading placed the liftoff at cos $\approx +0.07$; that reading was precisely the bulk-edge multiple-comparison artifact the null calibration exposed, and the corrected test relocates the significant excess to the high-similarity scale. The bulk of the discrepancy z remains the cone, per the ACG reference.) Operationally the corpus retains 83% of the ideal noise budget: a query may fade to cosine ≈ 0.25 against its target before an impostor is expected to win. The per-item field's low tail is real but shallow, and the merge tree finds one modest pocket: crowding here is *diffuse within subsets* rather than pocketed — the regime where global averages mislead optimistically while per-item counts still name the at-risk documents. The separability panel (labels provided, used only to condition the readout) shows high neighborhood purity (0.95) against near-zero silhouette (+0.04): subsets are locally coherent yet globally entangled within the cone — a reading no single number provides.

10 From Measurement to Gradient

The confusion kernel is differentiable, so the measurement layer doubles as a training-time regularizer with a stated guarantee.

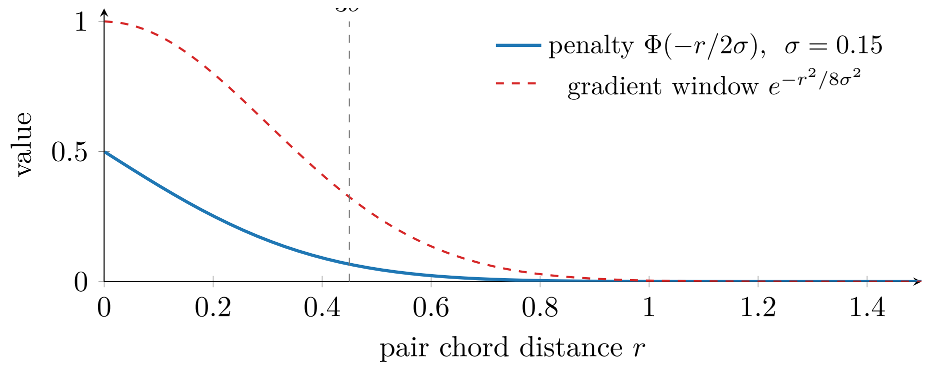

Figure 11: The confusion loss's gradient locality at a measured $\sigma = 0.15$ (the benchmark's σ^*): all gradient lives inside the confusable window; pairs beyond $\approx 3\sigma$ are exponentially untouched. "Improve resolution without disturbing dissimilarity" is a property of the kernel, not a hope.

10.1 Losses and the gradient-locality guarantee

For a batch $z_1, \ldots, z_B$ (normalized) and a *measured* target scale σ:

$$\mathcal{L}_{\text{conf}} = \underset{(i,j)\notin\mathcal{E}}{\text{mean}}\, \Phi\left(-\frac{\|z_i - z_j\|}{2\sigma}\right), \qquad \mathcal{L}_{\text{pres}} = \underset{i}{\text{mean}}\ \text{KL}\left(p_i^{\text{base}} \,\middle\|\, p_i\right),$$

where $\mathcal{E}$ excludes positive and guarded pairs and p_i (p_i^{base}) are temperature-softmax similarity rows under the tuned (frozen base) model. $\mathcal{L}_{\text{conf}}$ is the batch estimate of $C(\sigma)$; since $\partial\Phi(-r/2\sigma)/\partial r \propto e^{-r^2/8\sigma^2}$ (Figure 11), its gradient vanishes exponentially for pairs beyond $\approx 3\sigma$: far-pair (dissimilarity) structure *cannot* be disturbed, a property enforced in the test suite (a near pair receives $> 10^3\times$ the gradient of an orthogonal pair). $\mathcal{L}_{\text{pres}}$ is the neighbor-overlap comparison in differentiable form — the explicit statement of "do not rewrite who is similar to whom." (Implementation note: the chord computation floors $2 - 2c$ before the square root; at exactly zero the root's infinite derivative poisons masked entries with $0 \cdot \infty$ NaN gradients.)

10.2 Mining, protocol, and measured effect

Batch composition is driven by the measurements: sampling weights proportional to per-item collision counts (with a uniform floor so the bulk anchors the geometry), and negatives drafted from the *confusable window* — between the measured liftoff and the alignment scale of true pairs — under a false-negative guard: an anchor's top-m base-model neighbors are never negatives, because the window is precisely where unlabeled true relatives live. The protocol is staged: diagnose and take the cheapest licensed fix (cone $\Rightarrow$ linear post-processing; duplicates $\Rightarrow$ data repair; only genuine beyond-cone clumping licenses training); train with $\mathcal{L}_{\text{task}} + \lambda_c\mathcal{L}_{\text{conf}} + \lambda_p\mathcal{L}_{\text{pres}}$ at the measured σ; verify on a held-out reservoir never seen by mining, and never grade with the functional that was optimized.

Table 8 reports the worked end-to-end example. Flagged items' median expected collisions fall from 2.6 to 1.25 at held-out neighbor overlap 0.92; raising the preservation weight trades

Table 8: The worked training example, measured (a planted 300-item pocket in 4,000 items at $d = 64$; a linear adapter as the model stand-in; all verdicts from a held-out slice unseen by mining or batching). A linear adapter roughly halves flagged-item collisions at high structure preservation; we report this as the attainable ceiling of the *adapter* — which cannot unfold a tight pocket further without moving everything else, exactly the trade $\mathcal{L}_{\text{pres}}$ enforces — not of the method.

schedule	flagged median collisions	neighbor overlap w/ base
before	2.59	1.00
Adam, 10^{-2}, 600 steps, λ_p=0.3	1.25	0.92
Adam, 3×10^{-2}, 600 steps, λ_p=0.3	1.32	0.91
Adam, 3×10^{-2}, 1000 steps, λ_p=0.5	1.60	0.93

Table 9: The loop on the case-study corpus (200k measurement subset; verdicts from the held-out 20% only; flagged cohort = top 1% of items by base-model expected collisions, tracked at fixed $\sigma = 0.134$ across rounds). The misspecified first round is *rejected by measurement*; the corrected rounds are safe but marginal at every capacity.

round	σ^*	liftoff	z	pocket (size)	overlap	cohort med.
base	0.1337	0.769	−3565	0.085 (294)	1.00	2.58
linear adapter	0.1331	0.770	−3665	0.064 (17)	0.979	2.82
LoRA, mismatched refs	0.1324	0.758	−3775	0.091 (319)	0.924	3.01
LoRA, corrected	0.1339	0.763	−3545	0.082 (307)	0.973	2.51
full fine-tune	0.1338	0.763	−3552	0.083 (275)	0.965	2.53

recovery for overlap, exactly as designed. The guard matters empirically: without it, window mining drafts unlabeled true relatives as negatives — the classic silent failure of hard-negative mining.

10.3 Closing the loop on a production encoder

The synthetic example establishes the mechanism; the remaining question is whether the loop — measure, construct the objective from the measurements, train, re-embed, re-measure — operates on a production encoder and a real corpus, with the measurements adjudicating every round. We ran this loop on the case-study corpus (§9) and its 0.6-billion-parameter instruction-tuned encoder. A fixed, stratified 200,000-document measurement subset (proportional across the four subsets) was drawn once; 20% was held out and never touched by mining, sampling weights, or training. The objective is fully unsupervised: $\mathcal{L}_{\text{conf}}$ at the subset's measured $\sigma^* = 0.134$ under the false-negative guard, plus $\lambda_p\mathcal{L}_{\text{pres}}$ against frozen base-model references, with batches composed half from collision-weighted sampling and half from guarded confusable-window pairs (window = the measured liftoff cosine 0.769 to 0.98). Between rounds, a pre-registered licensing rule decides continuation: σ^* must not fall and held-out neighbor overlap must remain at least 0.90.

Table 9 summarizes the rounds; three findings follow. First, *the instrument rejects a misspecified round.* The initial LoRA round computed $\mathcal{L}_{\text{pres}}$ against stored embeddings of full documents while the trainer embedded token-truncated views, so the gradient was dominated by a length distillation; training losses fell while every global readout degraded (σ^* down, cohort collisions up 17%, the pocket enlarged). The licensing rule caught this from the held-out measurements alone. Second, with references matched to the trainer's own views and the confusable window placed in every batch, both LoRA and full fine-tuning are *safe but marginal*: no readout degrades, held-out overlap stays at 0.97, and the flagged cohort improves by 2–3% in the median — a small effect that is nonetheless *uniform*: 406/408 (LoRA) and 356/408 (full fine-tune) of held-out flagged items individually reduce their expected collisions (paired sign tests, $p < 10^{-100}$ and $p < 10^{-56}$). σ^* is measured as genuinely unchanged rather than noisily so: the paired difference across eight independent pair samples is +0.00007 against a per-sample standard deviation of 0.00012. All of this despite the full fine-tune moving roughly 40× the parameters of the LoRA round on a doubled pair batch. Third, the capacity ladder — linear adapter, LoRA, full fine-tune — shows that *capacity is not the binding constraint*: the linear adapter, which halved collisions on the synthetic single-pocket benchmark (Table 8), is itself rejected on the real corpus (σ^* falls and cohort collisions worsen 9% — a single linear map cannot serve the mixture of scales), while LoRA and the full fine-tune land within noise of one another. The corpus's dominant pocket is textual near-duplication, and the preservation term correctly refuses to separate items whose inputs are near-identical. This is the training-side confirmation of the staged protocol's routing rule: duplicate pathology is a data-repair problem, and no training objective that preserves base-model semantics can substitute for deduplication. Training at the measured scale is safe; it is not a shortcut around the data.

As a final check that the subset verdicts transfer, the full corpus was re-embedded with the accepted fine-tune and measured at full scale with the identical pipeline as the base case study. Every readout reproduces within measurement noise — σ^* 0.1227 against the base 0.1226, liftoff 0.817, effective rank 728, IsoScore 0.089, mean pair cosine 0.235, hubness skew 1.88, top pocket 198 items at prominence 0.090 against the base 204 at 0.090 — so the fine-tune that improves retrieval (below) is confirmed geometry-preserving at 10^6 documents, and the 200k held-out measurement is a faithful proxy for the full corpus.

10.4 Driving retrieval behavior

The loop above never touches a relevance judgment; judgments enter exactly once, after all rounds are frozen, to *score* the frozen predictions (the blind-then-score discipline of the validation plan). On the corpus's citation-grounded question-to-section retrieval set (1,000 held-out questions against a 35,173-section sub-corpus, exact search), the unsupervised full fine-tune moves recall@1 from 0.288 to 0.301 and MRR@10 from 0.467 to 0.475; the LoRA round is flat (0.286, 0.466). The recall gain does not reach significance at this query count: McNemar's exact test on the paired outcomes gives $p = 0.45$ (118 against 131 discordant queries), and the bootstrap interval on the MRR@10 difference, $[-0.008, +0.023]$, spans zero. The correct reading is therefore *retrieval-neutral-to-positive*: a per-item-significant geometry improvement whose aggregate task effect is bounded near zero, consistent with the duplication finding — most failures are not caused by the crowding the objective can legally repair. For context, a label-supervised fine-tune measured under the same harness reaches recall@1 0.355 — the supervised ceiling the unsupervised loop does not claim to reach. The per-item layer makes a sharper, item-level prediction: documents with high base-model expected collisions should be the ones that *fail at their own queries* (a judged-relevant document

Table 10: Dedup routing, scored blind on 1,000 frozen queries (fixed/broken = queries whose failure status at rank 10 changed). Confusable-window cleanup applies to foreign distractors; only true-duplicate merging applies within the formulaic corpus.

threshold	foreign distractors removed	target corpus merged
liftoff (0.82)	$+12/-0$ fixed, $p = 0.0005$	$+26/-57$, $p = 0.0009$ (harmful)
duplicates (0.95)	0/0 (no effect)	$+9/-2$, $p = 0.065$

outranked by non-relevant neighbors). Scored blind, base expected-collision counts predict failure participation at AUC 0.578 against 0.551 for the natural control (top-1 neighbor cosine) over 953 gold documents. The signal is significantly better than chance (bootstrap 95% interval $[0.536, 0.622]$) but the margin over the proximity control is not individually significant on this single corpus ($+0.028$, paired bootstrap interval $[-0.017, +0.073]$) — establishing the cross-corpus version of this comparison is precisely hypothesis H2 of the external validation plan.

In operational terms the instrument drives retrieval behavior through the readout-to-intervention map exercised end to end in this section: the anisotropy layer licenses linear post-processing; named near-duplicate pockets route to data repair before any training; genuine beyond-cone clumping licenses guarded training at the measured σ^*, verified on held-out measurements with a pre-registered continuation rule; per-item collision counts triage which documents to repair or re-chunk first; and σ^* itself sets the query-noise budget an application may spend.

10.5 The routing rule, executed and scored

The claim that duplicate pathology is a data problem was then tested directly: the repair the measurement prescribed was executed blind to all labels — near-duplicate groups identified purely from geometry (pairs above a stated cosine threshold, grouped by connectivity), each group reduced to one canonical member chosen by identifier order, and relevance judgments mapped through that reduction mechanically — then scored on the frozen evaluation. Two repair sites and two thresholds give a double dissociation (Table 10): removing *foreign* near-duplicates inside the confusable window (the liftoff scale) fixes 12 queries and breaks none (McNemar exact $p = 5 \times 10^{-4}$) — the significant retrieval gain that training could not deliver — while merging *within* the formulaic target corpus at that same scale is significantly harmful (57 broken), because the confusable window there contains legally distinct sections; only merging at the true-duplicate scale ($\cos \geq 0.95$) helps in-corpus. The instrument's scale structure — liftoff marks the onset of confusability, not the duplicate boundary — prescribes different repairs at different scales, and the held-out labels confirm each mapping in both directions.

10.6 Guardrails for supervised fine-tuning

The strongest test of the training layer reverses its role: rather than driving an unsupervised objective, the measurements *guard* a label-supervised one. A prior supervised fine-tune of the case-study encoder had gained first-rank retrieval while silently collapsing an untargeted multi-hop evaluation — the failure class $\mathcal{L}_{\text{pres}}$ and the false-negative guard exist to prevent.

We ran the identical supervised recipe (in-batch contrastive loss on 45,465 question–section pairs, two epochs) in two arms differing only in the guardrails: a *control* (task loss alone) and a *guarded* arm (task loss + $\lambda_p \mathcal{L}_{\text{pres}}$ against the frozen base on identical truncated views, with batch documents whose base-model cosine to the anchor's positive exceeds the measured liftoff masked out of the contrastive denominator).

The held-out measurement gate, read before any judgment, ranked the arms: the guarded model dominates on every label-free readout (neighbor overlap 0.650 against 0.537; σ^* 0.146 against 0.136, the base being 0.134; flagged-cohort collisions 0.47 against 2.10). The frozen evaluations then confirmed the prediction. On the targeted task the guardrails cost nothing (MRR@10 0.503 against 0.501; both arms far above the base 0.467). On the untargeted open evaluation the guarded arm wins everywhere: single-document MRR@10 0.597 against 0.563, and on the multi-hop metric where the historical regression lived, all-gold@10 0.339 against 0.223 — a 52% relative improvement over the control (1,496 held-out queries; the gap exceeds five standard errors). The control reproduces the historical collapse; the guard halves the damage while improving every other number. Two conclusions follow: the measurement-derived guardrails make supervised domain adaptation safer at no cost to its gains, and the instrument's label-free held-out gate selects the better model before any label is consulted.

11 External Validation Across Corpora and Encoders

The preceding sections validate the instrument internally and on one corpus. This section reports a first cross-corpus, cross-encoder slice of the external-validation design: four public retrieval collections with relevance judgments, in their BEIR form [45] — TREC-COVID [46] (171k biomedical documents, 50 densely judged topics), Quora duplicate questions (523k, duplicate annotations), Natural Questions [47] (2.7M encyclopedia passages), and the Shopping Queries / ESCI product-search collection [48] (1.2M products, graded judgments) — crossed with seven public encoders spanning 22M to 0.6B parameters, three output dimensionalities, and distinct training lineages: all-MiniLM-L6-v2 and all-mpnet-base-v2 [49], E5-base-v2 [50], BGE-large-en-v1.5 [51], Arctic-Embed-m-v1.5 [52], EmbeddingGemma-300m [53], and the 0.6B instruction-tuned encoder of the case study. This yields 28 corpus–encoder cells, plus the legal case study. Per cell, the corpus is embedded, the full readout set is computed and its hash recorded *before* any judgment file is opened, the per-document collision field is computed at the cell's measured σ^*, and only then are the queries run and scored (nDCG@10, recall@100, and per-relevant-document failure labels). One pipeline check: BGE-large's nDCG@10 on TREC-COVID reproduces its published benchmark value. Table 11 summarizes the grid's two findings, elaborated below.

The per-item claim is confirmed across corpora. In every cell whose outcome variable is informative, expected-collision counts predict which judged-relevant documents fail at their own queries: 7/7 encoders on duplicate questions (AUC 0.66–0.74), 7/7 on product search (0.57–0.69, including 0.69 for the case-study encoder despite an 84% base failure rate), 7/7 on open-domain QA (0.51–0.62), and the legal case study (0.58) — 22/22 positive cells (sign test $p \approx 2 \times 10^{-7}$). The biomedical collection is uninformative for this outcome by construction, not by counterexample: with 50 queries carrying hundreds of relevant documents each, 98% of judged documents exceed rank 10 and the failure label saturates entirely.

The corpus-level claim earns a narrow scope, not a blanket confirmation.

Table 11: The external grid at a glance (7 encoders per corpus; readouts frozen before judgments). ρ is the within-corpus Spearman rank correlation across encoders between the readout and nDCG@10; "control" is the same correlation for mean pair cosine. H2 AUC is the range, across encoders, of the per-document collision field's ability to predict which judged-relevant documents fail at their own queries (rank >10).

corpus	task type	$\rho(\sigma^*)$	control	H2 AUC	H2 cells >0.5
Quora	duplicate matching	+0.75	−0.79	0.66–0.74	7/7
ESCI	product search	−0.07	+0.32	0.57–0.69	7/7
Natural Questions	open-domain QA	−0.54	+0.54	0.51–0.62	7/7
TREC-COVID	biomedical	−0.32	+0.39	(saturated)	—
legal (case study)	—	—	—	0.58	1/1

Within-corpus, across encoders, the rank correlation of σ^* with nDCG@10 is +0.75 only on duplicate matching — the one task where relevance *is* geometric proximity — and there the naive mean-cosine control is *inverted* (−0.79). On product search the correlation is null (−0.07; graded shopping relevance mixes attribute matching with intent the geometry cannot see), and on open-domain QA and biomedical search it reverses (−0.54, −0.32): the smallest encoder attains the best geometry and the worst retrieval. The mechanism is the alignment–uniformity decomposition made empirical: retrieval quality factors into semantic alignment and geometric headroom; ambit measures only the second (it is unsupervised by construction and cannot see alignment), so σ^* ranks encoders only where alignment is essentially constant across them — and a semantically weak encoder can win geometry by spreading documents it does not understand. The operational reading, now measured rather than asserted: the corpus-level σ^* is a *budget* for comparative use — the same corpus under encoder variants, repair, or drift, as the guardrail gate used it — not a leaderboard; the externally validated cross-encoder readout is the per-item collision field above.

Ground truth for the scale semantics. On the duplicate-question collection, whose judgments annotate true duplicates, predicting "duplicate" from pair cosine alone yields precision that rises monotonically along the threshold ladder (0.29 at $\cos 0.80$, 0.93 at 0.95, 0.98 at 0.98 for the case-study encoder), while recall at the duplicate scale is low because annotated duplicates are mostly paraphrases living in the confusable window. This is item-level confirmation of the two-scale reading — confusable onset versus duplicate boundary — that the routing experiment (§10.5) turned into opposite repair prescriptions.

12 Limitations and Future Work

External validation. A first cross-corpus slice now exists (§11): the per-item claim is confirmed in 22/22 informative cells and the corpus-level claim received a measured scope law. The full preregistered study remains open — more collections (one large multi-hop corpus is deferred with its queue preserved), encoder-degradation ladders (H3), preregistration, and independent replication — and the scope law itself is a hypothesis to re-test: the alignment–uniformity split predicts that adding any alignment probe would restore ranking power on knowledge-limited tasks, but such a probe requires supervision the tool deliberately excludes. The confusion transform makes one falsifiable prediction its near-redundant sibling does not

(Table 4): duplicate pathology outranks diffuse collapse. **Query distributions.** Accepting an optional *unlabeled* query set would replace the corpus-as-queries assumption with a measured cross-set confusion, within the tool's supervision constraint. **Sharper aggregates.** The union bound is safe but loose where hubs live; Gaussian comparison inequalities may yield two-sided estimates. **Recoverability.** "Resolution headroom" — the fraction of measured crowding removable by a budgeted transform class, computed rather than predicted — would attribute measured crowding to the embedding model versus the data themselves. **Streaming structure.** The pair curve sketches trivially; the merge tree does not; a mergeable approximate hierarchy with prominence error bounds would extend the structural layer past the reservoir.

13 Conclusion

Crowding is a failure mode that embedded systems exhibit without overt errors, and it is exactly the property that global averages cannot see and binned displays cannot be trusted to measure. The instrument described here rests on one methodological commitment applied uniformly: *read continuous objects at every scale against stated nulls, and descend to named items.* The mathematics is classical — Ripley's curve, Stolarsky's identity, the distance to a measure, the condensed cluster tree, a Gaussian tail bound — and the contribution is their selection, calibration, and composition into a system with stated guarantees and stated scope, traced end to end through a single benchmark whose planted pathology every layer detects, localizes, prices, and (in the training layer) partially repairs. The negative results are reported alongside the positive ones because an instrument's reliability rests in part on a precise statement of its limits.

Appendix A Engineering Notes

The implementation is organized so that its asymptotic costs match the claims of §9: one streaming pass over the corpus, followed by statistics whose cost is independent of corpus size. This appendix records the implementation decisions behind the measured constants of Table 6.

Streaming scan. Ingestion is a single sequential pass over raw `.npy` files (memory-mapped) or sharded Parquet (chunked decode), in batches of 5×10^4 rows. Each batch of b rows updates the running mean, the norm moments, and the $d \times d$ second-moment scatter through one $X_b^\top X_b$ GEMM, and feeds a bounded reservoir sample; peak memory is $O(d^2 + \text{reservoir})$ irrespective of corpus size, and the covariance spectrum is extracted once at the end of the pass. An optional approximate mode terminates the pass after a prescribed number of rows and estimates the spectrum from that prefix; §9 documents the failure mode this introduces on sorted shards.

Blocked kernels at reservoir scale. No computation materializes an $m \times m$ matrix for reservoir size m. Top-k cosine values (the input to the distance-to-measure field) and collision counts are computed from $b \times m$ Gram tiles with partial-selection top-k (`argpartition`, linear per row, in place of a full sort). The mutual-reachability minimum spanning tree uses a vectorized Prim iteration that maintains a frontier-distance array: $O(m^2)$ arithmetic executed as full-array operations, $O(m)$ memory beyond the current tile. The structural layer additionally caps its inputs (6,000 points for the DTM field, 4,096 for the merge tree), which fixes the corpus-size-independent constants observed in Table 6.

Null sampling without vector synthesis. All isotropic-null quantities exploit the closed form for the pair cosine of independent uniform directions on S^{d-1}, $c = 2B - 1$ with $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$: null pair samples are drawn directly from this scalar law, so a 199-replicate rank envelope requires 199 scalar-array draws rather than 199 synthetic corpora. The anisotropy-conditioned reference uses the rotation invariance of the pair-cosine law under the angular central Gaussian — only the covariance spectrum enters — so reference pairs are generated from a small pool of spectrum-weighted Gaussian directions with no eigenvector computation. The Gaussian confusion law is evaluated with a vectorized rational approximation to erf (absolute error below 1.5×10^{-7}), which keeps the core free of dependencies beyond `numpy`.

One shared pair sample. A single random-pair cosine sample (default 2×10^5 pairs) is drawn once per corpus and reused by every pairwise statistic — $K(t)$, the envelope test, the discrepancy z, the confusion functional $C(\sigma)$, and σ^*. The continuous layer therefore costs one pass over a fixed-size array per statistic, and all reported statistics refer to the same measured sample.

Distributed training and pair-batch statistics. The encoder-loop trainer (§10.3) runs data-parallel: each rank embeds its own rows and the embeddings are gathered across ranks (the local slice retaining its gradient) before the pair losses, so pair statistics scale with world size at fixed per-device memory. This matters beyond throughput: the pair losses see only within-batch pairs, and small pair batches permit batch-level overfitting — the model separates the pairs it is shown while degrading out-of-batch structure, a failure the loop's held-out measurements catch. Gathered batches must be identically shaped on every rank; the batch builder emits a fixed count of distinct rows per rank for this reason.

Optional acceleration tiers. The core is numpy-only, with the heavy kernels routed through multi-threaded BLAS. An optional module provides the same three kernels — covariance accumulation, pair cosines, and exact blocked kNN — on torch devices (CPU, CUDA, or Apple mps, with automatic selection), transferring each chunk to the device as it streams; an approximate-neighbor path through FAISS is available where installed. These tiers change constants, not asymptotics; every number reported in this paper was produced by the numpy-only configuration (Appendix C).

Appendix B The Report Artifact

The toolkit delivers its measurements as a single self-contained, theme-adaptive HTML report with no external requests. We document its design here for completeness and reproducibility; *the artifact itself is engineering, not an academic contribution* — the contribution of this report is the measurement layer of §6–§7, which is exposed as library functions independent of any rendering. Two of the artifact's design rules are nonetheless direct consequences of the measurement theory and are worth stating. *Reference, not threshold*: every panel draws its null, because §2 established that absolute cosine is not self-interpreting. *Display versus measurement*: 2-D projections lay figures out but never carry a statistic, because §5 established that projected density is not corpus density. The remaining choices are presentation craft: a single accent color with meaning carried by position along annotated axes; figures ordered as a narrative (see it; how crowded, at what scale; who is affected; what structure it forms; why the space behaves so); and an in-place explainer on every panel stating its unstated context (null construction, subsample sizes, id provenance). One legibility finding from the artifact's development is recorded because it reinforces the display/measurement separation: a minimum-spanning-tree view drawn over the projection — even restricted to tighter-than-bulk bridges — remained too busy to read on real corpora and was demoted to a hidden option; the condensed tree of §6.4 carries the same object legibly as prominence bars.

Appendix C Test Environments

Two CPU-only Linux hosts, described by their properties. The *measurement host* (validation battery, scaling study): a 16-core/32-thread x86-64 workstation, 123 GB RAM, NVMe storage with $\approx 2\,$GB/s measured streaming reads, Python 3.13, numpy 2.5 with multi-threaded OpenBLAS, and the *numpy-only core install* of the artifact — deliberately, since that is the configuration the portability claim is about. The *case-study host*: a 6-core/12-thread mobile-class x86-64 machine with 27 GB RAM, exercising the optional tabular-IO path (sharded Parquet with decode overhead). Every statistic is computed by the released library (the training experiments additionally use `torch` on CPU); no experiment uses code outside the artifact; random seeds are fixed and stated per experiment.

Appendix D Formula Glossary

Every formula in the report is restated here in plain language: what it formulates, why it matters, how it is used, and what every symbol and operation means. Entries follow the order of first appearance. The first entry explains the shared notation that recurs throughout; each later entry still lists all of its own symbols.

How to read the notation (used throughout).

- n — the number of documents in the corpus. d — the number of coordinates in each embedding vector (the "dimension").
- i, j — indices naming individual documents, each running from 1 to n. Writing x_i means "the embedding vector of document number i"; a pair (i, j) names two different documents.
- $\Pr[\,\text{event}\,]$ — "the probability that the statement inside the brackets is true." A subscript names what is random: in $\Pr_{(i,j)}[\cdots]$ the random thing is the choice of pair — draw two distinct documents at random, and ask how often the bracketed statement holds. With no subscript, the randomness is stated in the entry (for the flip law it is the query noise).
- $\{\sigma : C(\sigma) \le 1\}$ — "the set of all values σ such that the condition after the colon holds"; $\max\{\cdots\}$ takes the largest member of a set.
- $\sum_{j \ne i}$ — add up the expression that follows, once for every allowed value of the index (j running over all documents except i). $\frac{1}{n^2}\sum_{i,j}$ — add over all n^2 ordered pairs, then divide by their count: an average over pairs.
- $\int f(r)\, dK(r)$ — an average of f weighted by how often each distance r occurs among pairs; "$dK(r)$" means "the fraction of pairs whose distance is (approximately) r."
- $\mathbb{E}$ — the expected value (average) over the stated randomness.
- log — the natural logarithm; e^x — the exponential function, its inverse.
- $\partial f/\partial r$ — the derivative of f with respect to r: how fast f changes when r changes a little. $\propto$ — "proportional to": equal up to a constant factor that does not depend on the variable under discussion.
- $\|v\|$ — the Euclidean length of a vector v (square root of the sum of its squared coordinates). $\langle u, v\rangle$ — the dot product of two vectors (multiply matching coordinates, add them up).
- Φ — the standard normal cumulative distribution function: $\Phi(u)$ is the probability that a standard Gaussian (bell-curve) random number falls below u. $\Phi(0) = 1/2$; $\Phi(-4) \approx 3 \times 10^{-5}$.

Unit vectors and cosine similarity: $c_{ij} = \langle x_i, x_j \rangle$, points on $\mathbb{S}^{d-1}$.

- **What it says:** Every document is represented as a length-one arrow in d-dimensional space, and the similarity of two documents is the cosine of the angle between their arrows.
- **Why it matters:** Cosine is the score retrieval systems actually rank by, so all geometry in the report is expressed in the same units the system uses.
- **How it is used:** Every statistic in the report is a function of these pairwise cosines.
- **Term by term:**
  - x_i — the embedding vector of document i, rescaled to length 1 (each of its d coordinates is a number; the whole vector is one point in d-dimensional space).
  - $\mathbb{S}^{d-1}$ — the unit sphere in d dimensions: the set of all vectors of length exactly 1. The superscript $d-1$ is the standard name for the sphere's surface (a globe in 3-D space has a 2-D surface); nothing in the report depends on this convention.
  - $\langle x_i, x_j \rangle$ — the dot product of the two vectors, which for length-one vectors equals the cosine of the angle between them.
  - c_{ij} — shorthand for that cosine between documents i and j: $+1$ means identical direction, 0 unrelated (perpendicular), -1 opposite.

Chord distance: $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$.

- **What it says:** The straight-line distance between two points on the sphere, computed directly from their cosine.
- **Why it matters:** The query-noise law below is exact in chord distance, not in cosine, so the report converts between the two constantly; this identity is the dictionary.
- **Term by term:**
  - $x_i - x_j$ — the vector pointing from document j's position to document i's.
  - $\|x_i - x_j\|$ — that vector's length: the ordinary straight-line distance between the two points.
  - r_{ij} — shorthand for that distance.
  - $\sqrt{2 - 2c_{ij}}$ — the same distance computed from the cosine alone. It follows by expanding the squared length: $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle = 1 + 1 - 2c_{ij}$.
  - Reference points: identical items have $r = 0$; perpendicular ones $r = \sqrt{2} \approx 1.41$; opposite ones $r = 2$.

The null cosine scale: coordinates $\sim \mathcal{N}(0, 1/d)$, typical cosine $\approx 1/\sqrt{d}$.

- **What it says:** If points are spread uniformly over a high-dimensional sphere, each coordinate behaves like a Gaussian random number with variance $1/d$, and the cosine between two random points is almost always within a few multiples of $1/\sqrt{d}$ of zero (at $d = 768$, about ± 0.036).

- **Why it matters:** This is the baseline against which "suspiciously similar" is judged: in high dimension, unrelated items are *very* nearly perpendicular, so even modest cosines are extremely surprising under the null.

- **Term by term:**

  - $\sim$ — "is distributed as": the quantity on the left behaves like a random draw from the distribution on the right.
  - $\mathcal{N}(0, 1/d)$ — the normal (bell-curve) distribution with mean 0 and variance $1/d$: values concentrate near zero, spread $\approx 1/\sqrt{d}$.
  - $1/\sqrt{d}$ — the typical size of a null cosine; it shrinks as dimension grows, which is the concentration phenomenon.

Pair-closeness curve: $K(t) = \Pr_{(i,j)}[\, c_{ij} \geq t\,]$.

- **What it says:** For every similarity threshold t simultaneously, the fraction of document pairs that are at least t-similar.

- **Why it matters:** It is the report's primary object: a complete, bin-free record of how much of the corpus sits at every closeness scale. Crowding appears as an excess of very close pairs — the curve "lifting off" above its reference at high t.

- **How it is used:** Estimated exactly by sorting a large random sample of pair cosines; compared against the two null curves below; the comparison is tested with the rank-envelope test.

- **Term by term:**

  - K — the name of the curve (after Ripley's K function in spatial statistics, of which this is the spherical, cosine-coordinate version). For each input t it returns one number between 0 and 1.
  - t — a similarity threshold, swept continuously from low to high; the curve is read at every t at once, which is what "no bins" means.
  - (i, j) — a pair of two distinct documents, chosen uniformly at random from the corpus.
  - $\Pr_{(i,j)}[\, c_{ij} \geq t\,]$ — the probability, over that random choice of pair, that the pair's cosine reaches the threshold — equivalently, simply the fraction of all pairs that do.

The uniform null for pair cosines: $c = 2B - 1$ with $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$.

- **What it says:** If points were spread perfectly evenly on the sphere, the cosine of a random pair would follow this exact, known distribution — a Beta distribution stretched from $[0, 1]$ to $[-1, 1]$.

- **Why it matters:** It gives the "perfectly spread corpus" reference curve in closed form: no simulation of fake corpora is needed (one draws random *numbers*, not random *vectors*), which is what makes 199-replicate null bands cheap.

- **Term by term:**
  - c — the pair cosine whose null distribution is being stated.
  - B — a random draw from the Beta distribution, which produces numbers between 0 and 1.
  - $\text{Beta}(a, a)$ with $a = (d - 1)/2$ — a two-parameter family of distributions on $[0, 1]$; with both parameters equal it is a symmetric bell around 1/2 that narrows as dimension grows (the concentration phenomenon again).
  - $2B - 1$ — rescales $[0, 1]$ onto $[-1, 1]$ so the result is a cosine: $B = 1/2$ (the typical draw) maps to $c = 0$, i.e. perpendicular.

The anisotropy-matched reference (ACG): $x = \sqrt{\lambda} \odot g \,/\, \|\sqrt{\lambda} \odot g\|$.

- **What it says:** A second reference corpus that copies the real corpus's overall *shape* — its elongation along some directions and flattening along others — but contains no clustering whatsoever.
- **Why it matters:** Learned embeddings almost always occupy a narrow cone (the "cone effect"), which inflates all cosines benignly. Comparing the data against this reference, instead of against the uniform null alone, separates crowding the cone explains (benign) from crowding it cannot (genuine pockets).
- **Term by term:**
  - g — a vector of d independent standard Gaussian random numbers: a direction chosen with no preference at all.
  - λ — the corpus's covariance eigenvalues: a list of d non-negative numbers, one per principal direction, each saying how much variance the corpus has along that direction. Together they describe the corpus's ellipsoidal shape.
  - $\sqrt{\lambda} \odot g$ — multiply each coordinate of g by the square root of the matching eigenvalue ($\odot$ is coordinate-wise multiplication); this stretches the isotropic random cloud into the corpus's ellipsoid.
  - dividing by $\|\cdots\|$ — rescale to length one, so the reference point lands on the sphere like the real data.
  - Only the eigenvalues enter — rotating a corpus changes none of its pairwise cosines, so the eigen*vectors* are irrelevant here.

The global rank-envelope test and the liftoff cosine.

- **What it says:** To decide whether the data curve $K(t)$ genuinely exceeds the null band — across *all* thresholds at once, not cherry-picking one — the test simulates many null curves, ranks the data curve among them at every threshold simultaneously, and returns one global p-value. The *liftoff cosine* is the smallest similarity at which the data curve exits the envelope, reported only when the global test rejects.

- **Why it matters:** Reading exceedance off a pointwise band at each threshold separately commits a multiple-comparison error — on pure noise it fires essentially always (measured: every one of 40 null corpora). The global test's false alarms match its nominal rate.

- **How it is used:** Every liftoff claim in the report is gated on this global p; extreme-rank-length (ERL) ordering breaks ties among simulated curves.

- **Term by term:**

  - envelope — the band spanned by the simulated null curves (here 199 of them, drawn via the Beta law above).
  - p-value — the probability, if the corpus were truly well-spread, of seeing a curve as extreme as the data's anywhere along its length. Small p (e.g. 0.005) means "a well-spread corpus essentially never looks like this." With 199 simulations the smallest attainable p is $1/200 = 0.005$.
  - ERL (extreme rank length) — the tie-breaking rule: when two curves are equally extreme at their worst point, the one that stays extreme over a longer stretch ranks as more extreme.
  - liftoff cosine — the onset similarity of genuine tight-pair structure: below it the data track the reference; above it they exceed the envelope.

Stolarsky invariance and the occupancy discrepancy z: cap discrepancy = const $-\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$.

- **What it says:** Take every possible "cap" on the sphere (every center, every radius), compare how many points each cap holds against how many it should hold, square the errors, and add them all up. Stolarsky's theorem says this total — which looks uncomputable — equals a constant minus the average distance between pairs of points.

- **Why it matters:** It is the exact completion of what binned occupancy displays approximate: *every* lattice at *every* scale simultaneously, with no lattice to choose. A corpus that under-spreads (crowds anywhere, at any scale) has a smaller average pairwise distance.

- **How it is used:** The report computes the mean pairwise chord and standardizes it against the uniform null, giving the occupancy discrepancy z. On the benchmark, uniform controls score $z \approx 0$ and the planted pocket scores $z = -47$.

- **Term by term:**

  - cap discrepancy — the total squared error between actual and ideal cap occupancy, accumulated over all cap centers and radii; the formal object on the left of the identity.
  - const — an explicit constant that depends only on n and d, not on where the points are; it plays no role in comparisons.

- $\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$ — the average chord distance over all ordered pairs of documents (sum over every pair, divide by the number of pairs).
- z — the reported score: the observed mean chord minus the null mean, divided by the null standard deviation, where the null mean and standard deviation come from redrawing the same number of pair cosines from the Beta law. $z = -47$ reads "the corpus's average distance sits 47 standard deviations of sampling noise below the well-spread value."

Distance to a measure: $\mathrm{DTM}_m(x_i) = \left(\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2\right)^{1/2}$ **with** $k = \lceil mn \rceil$**.**

- **What it says:** For each item: the typical distance it must reach to gather the nearest m-fraction of the corpus around it — a per-item crowding radius. Small radius = crowded; large radius = isolated.
- **Why it matters:** It descends from corpus-level statistics to *named items*, which is what makes findings actionable, and it is provably stable: outliers and small perturbations of the corpus move it only slightly (the Wasserstein-stability guarantee).
- **How it is used:** Computed for every reservoir item; its low tail (the most crowded items) is compared against the uniform corpus's band; the report lists the most-crowded and most-isolated items by identifier.
- **Term by term:**
  - $\mathrm{DTM}_m(x_i)$ — the function's name ("distance to the measure") applied to document i at mass fraction m; one number per document.
  - m — the mass fraction, the method's one knob (e.g. 0.02: gather 2% of the corpus); being a fraction makes it scale-free in corpus size.
  - $k = \lceil mn \rceil$ — that fraction converted to a count of neighbors; $\lceil \cdot \rceil$ rounds up to a whole number.
  - $r_{i,(j)}$ — the distance from item i to its jth nearest neighbor; the parenthesized subscript (j) means the distances have been sorted ascending, so (1) is the closest.
  - $\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2$ — the average of the squared distances to the k nearest neighbors.
  - $(\cdots)^{1/2}$ — the square root of that average: a root-mean-square radius, an average robust to any single neighbor.

The merge tree: mutual reachability, birth, death, prominence.

- **What it says:** Connect items by their shortest bridges (single linkage) under the *mutual-reachability* distance. As the connection scale grows, tight groups form and later dissolve into the bulk. A pocket's *birth* is the scale at which it first holds together, its *death* the scale at which it merges into something larger, and its *prominence* = death − birth is how long it exists as its own entity.

- **Why it matters:** Prominence replaces every flat clustering threshold: a pocket is reported not because it passed a cutoff but because it persists across a wide range of scales. A null-calibrated prominence floor (the 99th percentile of prominence produced by pure noise at matched size and dimension) suppresses spurious pockets.
- **Term by term:**
  - single linkage — build connections by repeatedly merging the two closest groups; the merge tree records the order and scale of every merge.
  - mutual reachability of a, b — $\max\{d(a,b), \mathrm{core}(a), \mathrm{core}(b)\}$: the largest of the three listed quantities, where $d(a,b)$ is the ordinary distance between the two points and $\mathrm{core}(x)$ is the distance from x to its own kth nearest neighbor. Taking the max means two points count as close only if each is also locally dense — a noise point far from everything cannot form a spurious bridge.
  - birth / death — the connection scales at which a group first assembles and at which it is absorbed into an older group (the *elder rule*: when two groups meet, the younger one dies).
  - prominence — death minus birth, in distance units: the width of the scale range over which the pocket is a separate entity.
  - excess of mass — the selection rule choosing, on each branch of the tree, the cluster that persists longest.

The query-noise model: $q = x + \sigma g$, $g \sim \mathcal{N}(0, I_d)$.

- **What it says:** A query aimed at item x lands nearby rather than exactly on it: x plus Gaussian noise of scale σ in every direction.
- **Why it matters:** It converts geometry into operational risk. Real queries paraphrase, truncate, and typo; σ is the abstract knob for all such imprecision, and every downstream quantity is a function of it.
- **Term by term:**
  - q — the query vector the system actually searches with.
  - x — the item the query "means": its intended target.
  - $g \sim \mathcal{N}(0, I_d)$ — a fresh standard Gaussian vector: each of the d coordinates is an independent bell-curve draw with mean 0 and variance 1 (I_d, the identity matrix, encodes "independent, unit variance, no preferred direction").
  - σ — the noise scale: how far, typically, the query lands from its target, in the same units as chord distance. $\sigma = 0$ means perfect queries; larger σ means sloppier ones.

The exact flip law: $\Pr[\langle q, y\rangle > \langle q, x\rangle] = \Phi(-r/2\sigma)$.

- **What it says:** The probability that a noisy query aimed at x scores a competitor y higher than x itself depends on nothing but the distance between x and y, through one evaluation of the normal tail. Exactly — in any dimension.

- **Why it matters:** It is the bridge between geometry and retrieval errors, and it explains why near-duplicates are catastrophic: as $r \to 0$ the probability approaches $\Phi(0) = 1/2$ — a duplicated document loses half its queries to its twin no matter how small the noise is.
- **How it is used:** Verified by direct simulation to Monte-Carlo precision (max deviation 9×10^{-4}); summed over competitors to give $C(\sigma)$; used as the training loss kernel.
- **Term by term:**
  - x, y — the query's intended target and one competing document, both unit vectors; q — the noisy query from the previous entry.
  - $\langle q, y\rangle$ — the dot product of query and competitor: exactly the similarity score retrieval ranks by. The bracketed statement $\langle q, y\rangle > \langle q, x\rangle$ reads "the competitor scores higher than the intended target," i.e. the retrieval errs.
  - $\Pr[\cdots]$ — the probability of that error over the randomness of the query noise g.
  - r — the chord distance between x and y; σ — the query-noise scale.
  - $\Phi(-r/2\sigma)$ — the normal tail evaluated at $-r/2\sigma$. Why this argument: the set of points scoring x and y equally is the perpendicular bisector of the segment joining them, which sits at distance $r/2$ from x; Gaussian noise of scale σ crosses a boundary $r/2$ away with probability $\Phi(-(r/2)/\sigma)$. Larger separation or smaller noise pushes the argument more negative and the error probability toward zero.

Expected collisions: $C(\sigma) = \sum_{j\neq i} \Phi(-r_{ij}/2\sigma) \approx (n-1) \int \Phi(-r/2\sigma)\, dK(r)$.

- **What it says:** For an average item, the expected number of wrong competitors that a noisy query would rank above it: add the flip probability over every competitor.
- **Why it matters:** By Boole's inequality (the probability that *any* of several events occurs is at most the sum of their individual probabilities), $C(\sigma)$ bounds the probability of *any* retrieval error, no matter how the competitors are correlated — so it is conservative: crowding can make it loose, never wrong.
- **Term by term:**
  - $C(\sigma)$ — the function's name: expected collision count at noise scale σ; one number per σ, increasing in σ.
  - i — the item being queried; j — each competing document in turn.
  - $\sum_{j\neq i} \Phi(-r_{ij}/2\sigma)$ — the flip probability against competitor j (previous entry), added up over all $n-1$ competitors.
  - $(n-1) \int \Phi(-r/2\sigma)\, dK(r)$ — the same sum computed from the pair-distance distribution alone: average the flip probability over the distribution of pair distances (that is the integral against dK), then multiply by the number of competitors. This is why a random pair *sample* suffices — no per-item computation is needed for the corpus-level curve.

Resolution bandwidth: $\sigma^* = \max\{\sigma : C(\sigma) \le 1\}$**.**

- **What it says:** The largest amount of query noise the corpus can absorb before, in expectation, one wrong item outranks the right one.
- **Why it matters:** It is the report's headline operational scalar: a noise *budget* in retrieval units. Comparing σ^* against the same corpus size spread uniformly (the ratio reported alongside it) says what fraction of the theoretically available budget the embedding retains.
- **Term by term:**
  - $\{\sigma : C(\sigma) \le 1\}$ — the set of all noise scales at which the expected collision count stays at or below one; max takes the largest such scale.
  - the threshold 1 — "one expected collision per item": the point where retrieval starts, on average, to lose. It is a convention (a tolerance), and the report states it wherever σ^* appears.
  - C is monotone — more noise always means more collisions — so this largest admissible σ is well defined and is found by bisection: repeatedly halve an interval known to contain the crossing.

Uniformity as a collision bound: $U_t = \log \mathbb{E}\, e^{-t\|x-y\|^2}$ **and** $C(\sigma) \le (n-1)\, e^{U_t}$ **at** $t = 1/(8\sigma^2)$**.**

- **What it says:** The widely used Wang–Isola "uniformity" loss — the log of the average of e^{-tr^2} over pairs — is, at the right temperature, exactly the logarithm of a Chernoff upper bound on the collision count $C(\sigma)$.
- **Why it matters:** It gives the popular loss operational units: its temperature knob t is not folklore but a choice of query-noise scale ($t = 2$ corresponds to $\sigma = 0.25$). It also explains when uniformity misleads — it is a bound, and the report measures where the bound's ordering and the exact functional's ordering disagree.
- **Term by term:**
  - U_t — the uniformity functional's value at temperature t (one number per t; more negative = more spread).
  - x, y — a random pair of documents; $\mathbb{E}$ — the average over that random pair.
  - $e^{-t\|x-y\|^2}$ — a Gaussian-shaped penalty: 1 when the pair coincides, decaying with squared distance at rate t (large t = only very close pairs are penalized).
  - log — the natural logarithm of the average, following the original definition.
  - $t = 1/(8\sigma^2)$ — the dictionary between temperature and noise scale; where the identity holds.
  - the proof, in one line — the normal tail bound $\Phi(-z) \le e^{-z^2/2}$ (a Chernoff bound) applied to $z = r/2\sigma$ turns each flip probability into $e^{-r^2/8\sigma^2}$, and summing gives $(n-1)\, e^{U_t}$.

The global suite: mean pair cosine, IsoScore, effective rank, hubness skew.

- **What it says:** Four whole-corpus summaries used as context and as baselines. Mean pair cosine: the average c_{ij} over random pairs (how coned the corpus is). IsoScore: how close the covariance is to perfectly isotropic — equal variance in all directions — on a 0-to-1 scale. Effective rank: exp of the entropy of the normalized eigenvalue distribution — roughly, how many directions carry meaningful variance (between 1 and d). Hubness skew: the skewness (asymmetry) of the count of how often each item appears in others' k-nearest-neighbor lists — heavy right skew means a few "hub" items intrude into everyone's neighborhoods.

- **Why it matters:** These are the statistics practitioners already consult. The report's point is measured, not rhetorical: they are global averages, and the detection-power study shows them blind to small pockets that the continuous layer detects at full power.

Training losses: $\mathcal{L}_{\text{conf}} = \text{mean}_{(i,j)\notin\mathcal{E}}\, \Phi\big(-\|z_i - z_j\|/2\sigma\big)$ and $\mathcal{L}_{\text{pres}} = \text{mean}_i\, \text{KL}\big(p_i^{\text{base}} \,\|\, p_i\big)$.

- **What it says:** Two differentiable objectives built from the measurements. The first is the batch version of the collision count: the average flip probability over batch pairs, at the *measured* noise scale, excluding a set of protected pairs. The second pins local semantics: for each anchor, the distribution "who in this batch is similar to me" under the model being trained must stay close to the same distribution under the frozen base model.

- **Why it matters:** $\mathcal{L}_{\text{conf}}$ spends gradient only where confusion actually lives, and $\mathcal{L}_{\text{pres}}$ prevents the repair from destroying meaning. Their balance (λ_p) is the loop's central tension: recovery against fidelity.

- **Term by term:**
  - $\mathcal{L}$ — a loss: a number training tries to make small; the subscripts name the two losses (confusion, preservation).
  - z_i — the embedding of batch item i under the model currently being trained (it changes every step; x_i is reserved for the fixed corpus vectors).
  - $\text{mean}_{(i,j)\notin\mathcal{E}}$ — average over all ordered batch pairs *except* those in the excluded set; $\notin$ means "not a member of."
  - $\mathcal{E}$ — the excluded pairs: known positives and the false-negative guard (each anchor's top-m base-model neighbors, which are likely unlabeled true relatives and must not be pushed apart).
  - $\Phi(-\|z_i - z_j\|/2\sigma)$ — the flip law applied to the batch pair at the measured scale $\sigma = \sigma^*$.
  - $\text{KL}(p \,\|\, q)$ — the Kullback–Leibler divergence: the standard measure of how much a distribution q misrepresents a reference distribution p; zero exactly when they agree, larger the more they differ. The double bar $\|$ is part of the notation and separates reference from candidate.

- p_i and p_i^{base} — for anchor i, the softmax over its cosines to the rest of the batch (exponentiate each cosine divided by a temperature of 0.05, then normalize to sum to one — low temperature sharpens the distribution toward the nearest neighbors), computed under the trained model and under the frozen base model respectively.

Gradient locality: $\partial\Phi(-r/2\sigma)/\partial r \propto e^{-r^2/8\sigma^2}$.

- **What it says:** The derivative of the flip probability with respect to pair distance dies off as a Gaussian in r: pairs farther apart than about 3σ receive exponentially negligible gradient.
- **Why it matters:** It is a structural safety guarantee: minimizing $\mathcal{L}_{\text{conf}}$ *cannot* disturb the corpus's far-pair (dissimilarity) structure, by construction rather than by tuning. The test suite enforces it (a near pair receives more than 10^3 times the gradient of an orthogonal pair).
- **Term by term:**
  - $\partial\Phi(-r/2\sigma)/\partial r$ — how fast the flip probability changes as the pair distance r changes: exactly the signal backpropagation sends to a pair during training.
  - $\propto$ — proportional to: equal up to a constant that does not involve r.
  - $e^{-r^2/8\sigma^2}$ — a Gaussian in r: near 1 for close pairs, and collapsing extremely fast once r exceeds a few multiples of σ — at $r = 3\sigma$ it is already below $e^{-9/8} \approx 0.32$ of its peak and falling as the *square* of the distance in the exponent.

Neighbor overlap at k.

- **What it says:** For each item, the fraction of its k nearest neighbors that two embeddings (base and tuned) have in common; the report uses $k = 10$ and averages over held-out items.
- **Why it matters:** It is the retrieval-relevant measure of "did the model change who is similar to whom" — comparing neighbor *identities*, which global similarity statistics can miss — and it is the loop's licensing criterion (≥ 0.90).
- **Term by term:**
  - k — the neighborhood size: how many nearest neighbors each item's list contains (10 here, matching a retrieval page).
  - overlap for one item — $|A \cap B|/k$, where A and B are the item's neighbor sets under the two embeddings, $\cap$ is the set intersection (members common to both), and $|\cdot|$ counts members: 1 means identical neighborhoods, 0 fully reshuffled.

Significance machinery of the loop section.

- **McNemar's exact test** asks whether a model change helped or hurt on *paired* outcomes: it looks only at the discordant queries (helped by one model, failed by the other — here 118 against 131) and asks whether that split is farther from 50:50 than coin flips would produce.

- **The paired sign test** does the same for per-item quantities: of 408 held-out flagged items, 406 improved — the probability of a split that lopsided under "no effect" is astronomically small ($p \approx 10^{-118}$), which is how a tiny median effect can still be unambiguous.

- **Bootstrap confidence intervals** quantify uncertainty without distributional assumptions: resample the data with replacement many times, recompute the statistic each time, and read the spread; an interval containing zero means the sign of the effect is not established.

- **AUC** (area under the ROC curve, equivalently the Mann–Whitney statistic) measures a predictor's ability to rank: the probability that a randomly chosen failing document scores higher on the predictor than a randomly chosen non-failing one — 0.5 is chance, 1.0 is perfect ordering.

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